

ON THE FIRST OCCURRENCE OF AN INTEGER AS A PREFIX SUM OF DIVISORS

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ABSTRACT. For a positive integer N , let $f(N)$ be the least positive integer whose increasing list of divisors has an initial segment with sum N , when such an integer exists. We prove that f is well-defined for all positive integers $N \neq 2, 5$. Next, answering a question of Erdős, we show that $f(N) = o(N)$ fails in a strong sense, whereas $\limsup_{N \rightarrow \infty} f(N)/N = \infty$. Arbitrarily small and arbitrarily large ratios $f(N)/N$ each occur on sets of positive lower density. Moreover, we obtain a doubly exponential upper bound for the lower tail and quantitative lower bounds for the upper tail.

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1. INTRODUCTION

(sec:introduction)

Write the positive divisors of an integer $n \geq 1$ as

$$1 = d_1(n) < d_2(n) < \cdots < d_{\tau(n)}(n) = n.$$

Here $\tau(n)$ is the number of positive divisors of n , and $\sigma(n) := \sum_{d|n} d$ denotes their sum. A positive integer N is *represented by a prefix of divisors* (or simply *represented*) if

$$N = d_1(n) + \cdots + d_k(n) \quad \text{for some } n \in \mathbb{N} \text{ and } 1 \leq k \leq \tau(n).$$

Let $\mathcal{R}$ be the set of represented positive integers, and define $f: \mathcal{R} \rightarrow \mathbb{N}$ by

$$f(N) := \min\{n \in \mathbb{N} : N = d_1(n) + \cdots + d_k(n) \text{ for some } 1 \leq k \leq \tau(n)\}$$

for every $N \in \mathcal{R}$.

The obvious inequality $f(\sigma(n)) \leq n$, together with the well-known unboundedness of $\sigma(n)/n$, gives $\liminf_{N \rightarrow \infty, N \in \mathcal{R}} f(N)/N = 0$; see a comment under [8]. Erdős asked whether $f(N) = o(N)$ holds as $N \rightarrow \infty$, whether this might at least hold for almost all N (in an unspecified sense), and whether $\limsup f(N)/N = \infty$. This problem appears in Guy's collection of open problems in number theory [10, Problem B2] and is catalogued as Problem 1054 on Bloom's website *Erdős problems* [2]. Its variant is also mentioned in Erdős and Graham's problem book [7, p. 93]; the convention there omits the divisor 1, which only results in an immaterial shift of the function f . The computed values of f are recorded in OEIS sequence A167485 [8], where it is also asked whether

$$\mathcal{R} = \mathbb{N} \setminus \{2, 5\}. \tag{1.1} \quad \boxed{\text{eq:exact-represen}}$$

These questions are not easy to answer because the prefix sums of divisors $d_1(n) + \cdots + d_k(n)$ behave quite erratically. Nevertheless, we successfully rigorously address all of them by providing nontrivial information on the limiting distribution of the ratios $f(N)/N$. Moreover, we give a computer-assisted proof of the property (1.1).

1.1. Quantitative estimates for tails. Let $\#S$ denote the cardinality of a finite set S . For $S \subseteq \mathbb{N}$, write

$$\underline{d}(S) := \liminf_{X \rightarrow \infty} \frac{\#(S \cap [1, X])}{X}, \quad \overline{d}(S) := \limsup_{X \rightarrow \infty} \frac{\#(S \cap [1, X])}{X}$$

respectively for *lower* and *upper densities* of S . When these densities agree, their common value is denoted by $d(S)$. Throughout, $\log$ denotes the natural logarithm, and logarithmic expressions are used only where they are defined.

The first result bounds the frequency of small ratios $f(N)/N$ uniformly in the cutoff.

(thm:small-upper)

Theorem 1.1. *There is an absolute constant $c > 0$ such that, for all sufficiently small $\delta > 0$ and every $X \geq 1$,*

$$\#\{N \leq X : N \in \mathcal{R}, f(N) \leq \delta N\} \ll X \exp\{-\exp((1/\delta)^c)\}.$$

In particular, for every fixed $M > 0$ this counting function is $O_M(\delta^M X)$, uniformly in X .

An immediate asymptotic consequence of Theorem 1.1 is

$$\overline{d}(\{N \in \mathcal{R} : f(N) \leq \delta N\}) \ll \exp\{-\exp((1/\delta)^c)\}$$

for all sufficiently small $\delta > 0$. This fact already strongly refutes Erdős's intuition that $f(N)$ is typically much smaller than N .

Next, arbitrarily small ratios occur on sets of positive lower density.

(thm:small-values) **Theorem 1.2.** *For every $\delta > 0$ there is $c_\delta > 0$ such that*

$$\#\{N \leq X : N \in \mathcal{R}, f(N) \leq \delta N\} \geq c_\delta X$$

for all sufficiently large X .

This time, the asymptotic variant of Theorem 1.2 reads

$$\bar{d}(\{N \in \mathcal{R} : f(N) \leq \delta N\}) > 0$$

for every $\delta > 0$.

Regarding large ratios $f(N)/N$, our estimate decays more slowly than every negative power of the threshold.

(n:almost-log-tail) **Theorem 1.3.** *As $T \rightarrow \infty$,*

$$\underline{d}(\{N \in \mathcal{R} : N \text{ squarefree}, f(N) > TN\}) \geq \left(\frac{e^{-\gamma}}{2} - o(1) \right) \frac{\log \log \log T}{(\log T)(\log \log T)}. \quad (1.2) \quad \text{eq:almost-log-tail}$$

Consequently, $\{N \in \mathcal{R} : f(N) > TN\}$ has positive lower density for every fixed $T > 0$, and

$$\limsup_{\substack{N \rightarrow \infty \\ N \in \mathcal{R}}} \frac{f(N)}{N} = \infty.$$

Theorem 1.3 also settles the last part of Erdős's question in the affirmative. For every fixed $T > 0$ the set $\{N \in \mathcal{R} : f(N) > TN\}$ has positive lower density, so which is in an even sharper contrast with Erdős's conjecture $f(N) = o(N)$. It remains possible that the density of $\{N : f(N) > TN\}$ tends to zero as $T \rightarrow \infty$; Section 6 shows that this is equivalent to a certain coverage statement for bounded cofactors, and leaves it open.

The final theorem of this type allows the threshold to grow alongside with N , as opposed to fixing δ or T as before.

(thm:subexp-growth) **Theorem 1.4.** *For every fixed $\eta > 0$ there is $c_\eta > 0$ such that, for all sufficiently large X ,*

$$\# \left\{ \frac{X}{2} < N \leq X : N \in \mathcal{R}, N \text{ squarefree}, \frac{f(N)}{N} > \exp \left[\left(\frac{1}{\sqrt{2}} - \eta \right) \sqrt{(\log \log \log N)(\log \log \log \log N)} \right] \right\} \geq c_\eta \frac{X}{\log \log \log X}. \quad (1.3) \quad \text{eq:subexp-growth}$$

For every fixed $s > 0$ and $\eta > 0$, we have

$$\frac{1}{X} \sum_{\substack{N \leq X \\ N \in \mathcal{R}}} \left(\frac{f(N)}{N} \right)^s \geq \exp \left[\left(\frac{s}{\sqrt{2}} - \eta \right) \sqrt{(\log \log \log X)(\log \log \log \log X)} \right] \quad (1.4) \quad \text{eq:positive-momen}$$

for all sufficiently large X .

1.2. Hypothetical limiting distribution. The tail estimates from Theorems 1.1–1.3 suggest studying the empirical probability measures

$$\nu_X := \frac{1}{R(X)} \sum_{\substack{N \leq X \\ N \in \mathcal{R}}} \delta_{f(N)/N}, \quad R(X) := \#(\mathcal{R} \cap [1, X]), \quad X \geq 1, \quad (1.5) \quad \text{eq:empirical-meas}$$

where δ_x is unit mass at $x \in (0, \infty)$. Since we will see that $R(X) = \lfloor X \rfloor - 2$ for $X \geq 5$, the normalization by $R(X)$ is asymptotically equivalent to the one by X . Investigation of weak convergence of the measures (1.5) can be motivated by the classical analogous results for the sequences $\varphi(N)/N$, see [17], or $\sigma(N)/N$, see [4]. We find it plausible that the limiting distribution also exists in the case of the sequence $f(N)/N$, but this might be out of reach of the currently available techniques. Still, the present paper attempts to provide some partial results and supporting evidence.

For a factor d of a number $n = ed$, the integer e is called its *cofactor*. Define

$$F_e(d) := \sum_{\substack{r|ed \\ r \leq d}} r, \quad G_E := \bigcup_{1 \leq e \leq E} F_e(\mathbb{N}).$$

We prove that the family of probability measures $\{\nu_X : X \geq 1\}$ is tight on $[0, \infty)$ if and only if

$$\lim_{E \rightarrow \infty} \underline{d}(G_E) = 1,$$

but this “coverage assertion” remains a conjecture.

The distributional conclusions go substantially beyond tail bounds. There is one fixed σ -compact Lebesgue-null set $\mathcal{Z} \subset (0, \infty)$ carrying the finite part of every compactified subsequential law. Every such law has no finite atoms, has 0 in its support, and has infinite positive moments and infinite logarithmic mean. In particular, any subsequential probability limit on $[0, \infty)$ is singular continuous. These conclusions follow from an explicit limiting measure for all representation witnesses and the singularity of the distribution of $\sigma(n)/n$.

We also identify the remaining difficulty for the problem of weak convergence of (1.5). For $t > 0$, let $r_t^*(N)$ count representations $N = F_e(d)$ with $e \geq 2$ and $ed \leq tN$. Its normalized first moment has a continuous limit $W^*(t)$, and

$$\nu_X([0, t]) = W^*(t) - \frac{1}{X} \sum_{N \leq X} (r_t^*(N) - 1)_+ + o(1).$$

Thus, existence of the threshold density is equivalent to convergence of the displayed surplus term. This surplus has lower limit strictly larger than $1/9$ for every $t \geq 2$. It cannot be discarded and it causes complications.

1.3. Proof strategies. For small ratios $f(N)/N$, the main inspiration for the proofs is Tao’s unpublished comment under [2]. It implies a weaker variant of Theorem 1.1, with $\delta^2 X$ on the right-hand side, and it already disproves $f(N) = o(N)$. Following his suggestion, we prove an estimate for high moments of all prefix sums, which eventually gives Theorem 1.1. To obtain positive density, we use sums of proper divisors of a prescribed abundant family. The proof adapts the regularity decomposition and collision sieve of Luca and Pomerance [12], with Pollack’s control of successive aliquot ratios [14]. The fixed abundant factor makes the represented ratio arbitrarily small, while the second moment estimate ensures that a positive proportion of the images are distinct.

For large ratios, the cofactor separates the problem into a moment bound for large e and an arithmetic obstruction for bounded e . Typically $\sigma(n)$ is divisible by a prescribed fixed modulus. If the target is rough, the complementary divisor sum therefore has to be rough as well. Its exact forced modulus makes a large divisor of a finite least common multiple unavoidable. A Rankin estimate for the resulting divisor tail yields Theorem 1.3, while a uniform divisor-sum normality estimate and a different choice of parameters give Theorem 1.4. We use a single cofactor parametrization for the large value estimates and the witness measure approach.

For the limiting measures, fix a cofactor and a residue class modulo $\text{lcm}(1, \dots, e)$. The reciprocal prefix sum is then $\sigma(n)/n$ minus a fixed rational number. Davenport’s theorem in arithmetic progressions [15] and the large-cofactor moment bound give a convergent witness measure. On $(0, \infty)$, every subsequential ratio law is dominated by that witness measure. Erdős’s singularity theorem [6] consequently transfers to every finite subsequential law. This argument determines regularity of possible laws without assuming either tightness or uniqueness.

1.4. Notation. We use $s(n) := \sigma(n) - n$ for the sum of the proper divisors, φ for Euler's totient function, ζ for the Riemann zeta function, and γ for Euler's constant. For integers $a_1, \dots, a_k$, we write $\text{lcm}(a_1, \dots, a_k)$ for their least common multiple and $\text{gcd}(a, b)$ for the greatest common divisor of a and b . We write $v_p(n)$ for the exponent of a prime p in n , and $p^a \parallel n$ for exact divisibility. The largest and least prime factors of n are $P^+(n)$ and $P^-(n)$, with $P^+(1) = 1$ and $P^-(1) = \infty$. An integer is *y-smooth* if $P^+(n) \leq y$, and *y-rough* if $P^-(n) > y$. For real $y \geq 1$, we use the *primorial* notation

$$y\# := \prod_{\substack{p \leq y \\ p \text{ prime}}} p,$$

where each prime not exceeding y occurs exactly once. The empty product is 1; in particular, $y\# = 1$ for $1 \leq y < 2$. For real $y, u \geq 1$, set

$$\Delta(y) := \frac{\varphi(y\#)}{y\#} = \prod_{\substack{p \leq y \\ p \text{ prime}}} \left(1 - \frac{1}{p}\right), \quad \Lambda(u) := \text{lcm}(1, \dots, \lfloor u \rfloor).$$

Thus $\Delta(y)$ is the natural density of integers coprime to $y\#$, and $\Lambda(u)$ is the least common multiple of all positive integers not exceeding u . The symbols $\pi(x)$ and $\pi(x; q, a)$ count primes and primes in a reduced residue class, respectively, and $\psi(x) := \sum_{p^a \leq x} \log p$. We use $u_+ := \max\{u, 0\}$ for the positive part of a real number. All implied constants are absolute unless their dependence is indicated. The limits involving a fixed threshold always take $X \rightarrow \infty$ first. Finally, for $j \geq 0$ we write

$$\sigma_j(n) := \begin{cases} d_1(n) + \dots + d_{\tau(n)-j}(n), & 0 \leq j < \tau(n), \\ 0, & j \geq \tau(n) \end{cases}$$

for the *prefix sum* of all but j largest divisors of n . We interpret σ_0 as σ .

2. ARITHMETIC PRELIMINARIES

tic-preliminaries)

We use the cofactor functions and prime-product notation introduced in Section 1. Define the complementary reciprocal sum

$$g_e(n) = \sum_{\substack{r|n \\ r \geq e}} \frac{1}{r}.$$

Since the final divisor occurs in its own prefix,

$$F_e(d) \geq d, \quad F_e(d) \leq X \implies ed \leq eX.$$

Mertens' theorem and Chebyshev's estimates give

$$\Delta(y) \sim \frac{e^{-\gamma}}{\log y}, \quad \log(y\#) \ll y, \quad \log \Lambda(u) = \psi(u) \leq C_\Lambda u;$$

see, for example, [18].

Reflecting the divisor list expresses each prefix sum in terms of the complementary reciprocal sum.

Lemma 2.1. *For every $e, d \geq 1$,*

$$F_e(d) = ed g_e(ed). \tag{2.1} \boxed{\text{eq:reflection}}$$

Proof. The involution $r \mapsto ed/r$ exchanges the divisors $r \geq e$ of ed with its divisors at most d . $\square$

The omitted divisors force a congruence whose modulus depends on only finitely much divisibility information. Fix $e \geq 2$ and $d \geq 1$, put $n = ed$, and let

$$D = \{j : 1 \leq j < e, j \mid n\}$$

be the set of divisors of n below e . Put

$$M = \text{lcm}(D), \quad C = \sum_{j \in D} \frac{M}{j}, \quad g = \gcd(e, M), \quad K_{e,d} = \frac{e}{g} C.$$

Since $1 \in D$, these are positive integers. In fact $K_{e,d} \geq 2$: if $e/g > 1$ this is immediate, while $e/g = 1$ implies $C \geq M \geq e \geq 2$. For $1 \leq j < e$ write $h = \gcd(j, e)$. Then

$$j \mid ed \iff \frac{j}{h} \mid \frac{e}{h} d \iff \frac{j}{h} \mid d,$$

because j/h and e/h are coprime. Thus D depends only on the residue of d modulo

$$\text{lcm}(\{j/\gcd(j, e) : 1 \leq j < e\}).$$

In particular, for fixed e the integer $K_{e,d}$ takes only finitely many values.

lem:fm-modulus **Lemma 2.2.** *With the notation above,*

$$F_e(d) = \sigma(n) - \frac{n}{M} C, \tag{2.2} \text{eq:fm-reflection}$$

the integer M/g divides d , and

$$F_e(d) \equiv \sigma(n) \pmod{K_{e,d}}. \tag{2.3} \text{eq:fm-congruence}$$

Moreover $K_{e,d}$ is the largest modulus for which (2.3) holds for every d with the same set D .

Proof. The map $r \mapsto n/r$ is an involution of the divisors of n , and $r > d$ exactly when $n/r < e$. Hence

$$\sigma(n) - F_e(d) = \sum_{\substack{r \mid n \\ r > d}} r = \sum_{j \in D} \frac{n}{j} = \frac{n}{M} \sum_{j \in D} \frac{M}{j} = \frac{n}{M} C.$$

Every $j \in D$ divides n , so $M \mid n$ and n/M is an integer. This proves (2.2). Since $g = \gcd(e, M)$, the integers M/g and e/g are coprime. From $M \mid ed$ we get $(M/g) \mid (e/g)d$, hence $(M/g) \mid d$. Therefore

$$\frac{n}{M} = \frac{ed}{M} = \frac{e}{g} \cdot \frac{d}{M/g}$$

is e/g times an integer, and by (2.2) the difference $\sigma(n) - F_e(d)$ is a multiple of $(e/g)C = K_{e,d}$. For the last assertion take $d_0 = M/g$, so that $ed_0 = \text{lcm}(e, M)$. Every $j \in D$ divides M , hence ed_0 . Conversely ed_0 divides ed , so every divisor of ed_0 below e lies in D . Thus d_0 has the same set D , and at d_0 the difference $(ed_0/M)C$ equals $(e/g)C$ exactly. Any modulus valid for every d with this set D divides that difference, hence divides $K_{e,d}$. $\square$

If $P^-(d) \geq e$, then D is the set of divisors of e below e . Indeed, for $j < e$ the integer $j/\gcd(j, e)$ is below $e \leq P^-(d)$, so it divides d only if it equals 1, that is, only if $j \mid e$. In this case $M \mid e$, $g = M$, and

$$K_{e,d} = \sum_{\substack{j \mid e \\ j < e}} \frac{e}{j} = \sigma(e) - 1.$$

So on e -rough inputs, $F_e(d) \equiv \sigma(ed) \pmod{\sigma(e) - 1}$.

The sum of divisors is divisible by any fixed positive integer for almost all inputs.

modulus-normality **Lemma 2.3.** *For every fixed positive integer V ,*

$$\#\{n \leq Y : V \nmid \sigma(n)\} = o_V(Y) \quad (Y \rightarrow \infty).$$

Proof. It suffices to prove the assertion when $V = q$ is a prime power. For every prime $r \equiv -1 \pmod{q}$, the condition $r \parallel n$ implies

$$q \mid r + 1 = \sigma(r) \mid \sigma(n).$$

Therefore, if $q \nmid \sigma(n)$, none of these primes divides n exactly once. For any finite set $\mathcal{P}$ of such primes, the Chinese remainder theorem gives

$$\limsup_{Y \rightarrow \infty} \frac{1}{Y} \#\{n \leq Y : q \nmid \sigma(n)\} \leq \prod_{r \in \mathcal{P}} \left(1 - \frac{1}{r} + \frac{1}{r^2}\right).$$

The reciprocal sum of the primes in the reduced residue class $-1 \pmod{q}$ diverges. Letting $\mathcal{P}$ increase through this class makes the displayed product tend to zero. Factoring a general fixed V into its prime-power divisors and applying the union bound completes the proof. $\square$

This divisibility property forces the range of the full divisor sum to have density zero.

$\langle \text{sigma-range-zero} \rangle$ **Lemma 2.4.** *The set $\sigma(\mathbb{N}) = \{\sigma(n) : n \geq 1\}$ has asymptotic density zero.*

Proof. If $N = \sigma(n) \leq X$, then $n \leq X$. For every fixed integer $q \geq 2$, the values divisible by q number at most X/q , while all remaining values arise from $o_q(X)$ exceptional sources by Lemma 2.3. Hence

$$\bar{d}(\sigma(\mathbb{N})) \leq \frac{1}{q}.$$

Letting $q \rightarrow \infty$ proves the claim. $\square$

For a prime q and a real number $y \geq 1$, define

$$B_q(y) = \#\{n \leq y : q \nmid \sigma(n)\}.$$

The following estimate follows from a mean-value theorem for nonnegative multiplicative functions and the Siegel–Walfisz theorem.

$\langle \text{lem:sigma-rate} \rangle$ **Lemma 2.5.** *There are absolute constants $c_0, c_1 > 0$ such that, for all sufficiently large y and every prime*

$$3 \leq q \leq c_0 \frac{\log \log y}{\log \log \log y},$$

one has

$$B_q(y) \ll y \exp\left(-c_1 \frac{\log \log y}{q}\right).$$

Also $B_2(y) \ll \sqrt{y}$.

Proof. For prime powers define the multiplicative function h_q by

$$h_q(p^\nu) := \begin{cases} 0, & p \equiv -1 \pmod{q} \text{ and } \nu = 1, \\ 1, & \text{otherwise.} \end{cases}$$

If $p \equiv -1 \pmod{q}$ and $v_p(n) = 1$, then $q \mid \sigma(p) \mid \sigma(n)$. Hence $\mathbf{1}_{q \nmid \sigma(n)} \leq h_q(n)$. Since $0 \leq h_q(p^\nu) \leq 1$, the Halberstam–Richert mean-value bound, in the form [14, Lemma 2.4], applies with $\lambda_1 = \lambda_2 = 1$. It gives, with an absolute implied constant,

$$\begin{aligned} B_q(y) &\leq \sum_{n \leq y} h_q(n) \\ &\ll y \exp\left(\sum_{p \leq y} \frac{h_q(p) - 1}{p}\right) \\ &= y \exp\left(-\sum_{\substack{p \leq y \\ p \equiv -1 \pmod{q}}} \frac{1}{p}\right). \end{aligned}$$

Choose $z = \exp(Cq^2)$ with C sufficiently large. After decreasing c_0 if necessary, the stated range for q ensures that $z \leq y$ for all sufficiently large y . By the Siegel–Walfisz theorem [18], uniformly for $t \geq z$,

$$\pi(2t; q, -1) - \pi(t; q, -1) \gg \frac{t}{q \log t}.$$

Summing over dyadic intervals from z to y therefore gives

$$\sum_{\substack{z < p \leq y \\ p \equiv -1 \pmod{q}}} \frac{1}{p} \gg \frac{1}{q} \log \frac{\log y}{\log z} = \frac{1}{q} (\log \log y - 2 \log q + O(1)).$$

In the stated range the last parenthesis is at least $\frac{1}{2} \log \log y$ for large y , proving the first assertion. Finally, $\sigma(n)$ is odd exactly when n is a square or twice a square, so $B_2(y) \ll \sqrt{y}$. $\square$

The next moment estimate extends the argument of Tao [2], which can be viewed as the first moment approach. Keeping the condition $e \mid n$ in the moment expansion gives the decay E^{1-k} needed below.

$\langle \text{lem:moment} \rangle$ **Lemma 2.6.** *For every integer $k \geq 2$ there is a constant C_k satisfying*

$$\log C_k \leq 2^{k+1} \log(2k+2) \tag{2.4} \text{eq:Ck-growth}$$

such that, for all $E, Z \geq 1$,

$$\sum_{n \leq Z} \sum_{\substack{e \mid n \\ e > E}} g_e(n)^k \leq C_k Z E^{1-k}. \tag{2.5} \text{eq:moment}$$

Consequently, for all real $A, X \geq 1$, the number of integers $N \leq X$ for which there are $e, d \in \mathbb{N}$ satisfying $N = F_e(d)$, $e > E$, and $ed \leq AN$ is at most

$$C_k A^{k+1} E^{1-k} X. \tag{2.6} \text{eq:large-count}$$

Proof. Expanding the k th power and counting common multiples gives

$$\begin{aligned} & \sum_{n \leq Z} \sum_{\substack{e \mid n \\ e > E}} g_e(n)^k \\ &= \sum_{e > E} \sum_{r_1, \dots, r_k \geq e} \frac{1}{r_1 \cdots r_k} \left\lfloor \frac{Z}{\text{lcm}(e, r_1, \dots, r_k)} \right\rfloor \\ &\leq Z \sum_{e > E} \sum_{r_1, \dots, r_k \geq e} \frac{1}{r_1 \cdots r_k \text{lcm}(e, r_1, \dots, r_k)}. \end{aligned}$$

Set $x_0 = e$, $x_i = r_i$, and $R = x_0 \cdots x_k$. Since $r_1 \cdots r_k \geq e^k$ and $e > E$,

$$\frac{R^{1/(k+1)}}{r_1 \cdots r_k} \leq e^{1-k} \leq E^{1-k}.$$

Consequently,

$$\frac{1}{r_1 \cdots r_k \text{lcm}(e, r_1, \dots, r_k)} \leq E^{1-k} \frac{R^{-1/(k+1)}}{\text{lcm}(x_0, \dots, x_k)}.$$

For a prime p and $t \geq 1$, let $S_{p,t} = \{i : v_p(x_i) \geq t\}$, and for every nonempty $S \subseteq \{0, \dots, k\}$ put

$$y_S := \prod_p p^{\#\{t \geq 1 : S_{p,t} = S\}}.$$

Then

$$R = \prod_S y_S^{|S|}, \quad \text{lcm}(x_0, \dots, x_k) = \prod_S y_S.$$

Dropping the nesting restrictions on the sets $S_{p,t}$ only enlarges the sum, and hence

$$\begin{aligned} \sum_{x_0, \dots, x_k \geq 1} \frac{R^{-1/(k+1)}}{\text{lcm}(x_0, \dots, x_k)} &\leq \prod_{\emptyset \neq S \subseteq \{0, \dots, k\}} \sum_{y \geq 1} y^{-1-|S|/(k+1)} \\ &= \prod_{j=1}^{k+1} \zeta \left(1 + \frac{j}{k+1} \right)^{\binom{k+1}{j}} =: C_k. \end{aligned}$$

This proves (2.5). Since $\zeta(1+u) \leq 1+u^{-1}$ for $u > 0$, summing the binomial exponents gives (2.4).

For (2.6), put $n = ed$. By (2.1), $N = ng_e(n)$. Such a pair therefore satisfies $n \leq AX$ and $g_e(n) = N/n \geq 1/A$. The number of distinct values N is at most the number of such pairs (n, e) , which by Markov's inequality and (2.5) is at most

$$A^k \sum_{n \leq AX} \sum_{\substack{e|n \\ e > E}} g_e(n)^k \leq C_k A^{k+1} E^{1-k} X. \quad \square$$

3. REPRESENTABILITY

We first separate the analytic representability input from the finite computations needed for the exact classification. This distinction is useful because the large-value estimates require only the former.

The analytic input shows that almost every odd integer is represented, with a quantitative bound for the exceptional set.

-representability) **Lemma 3.1.** *There is an absolute $c > 0$ such that, for $X \geq 3$,*

$$\#\{n \leq X : n \text{ odd}, n \notin \mathcal{R}\} \ll X^{1-c} + \frac{X}{\log X}.$$

In particular, this exceptional set has size $o(X/\log \log \log X)$ as $X \rightarrow \infty$.

Proof. The theorem of Montgomery and Vaughan [13] asserts that at most $O(X^{1-c})$ even integers at most X fail to be sums of two primes. Discard also integers of the form $2p$, of which there are $O(X/\log X)$. Each remaining even integer has a representation $p + q$ with distinct primes $p < q$. The divisors of pq are $1, p, q, pq$, so $1 + p + q$ is a divisor-prefix sum. Translating by 1 proves the assertion. $\square$

We now prove the exact representability theorem by combining elementary constructions, finite verification, and analytic estimates. We identify the finite checks precisely and use a short rational estimate to extend the verified range.

-representability) **Theorem 3.2.** *The represented positive integers are precisely*

$$\mathcal{R} = \mathbb{N} \setminus \{2, 5\}.$$

In particular, $f(N)$ is defined for every $N \geq 6$.

3.1. Computational verification for finitely many integers. Subset sums of primes in a suitable interval become divisor-prefix sums after adding 1.

ture-prime-window) **Lemma 3.3.** *Let $M > 1$, and suppose that $M < p_1 < \dots < p_t \leq Y < M^2$ are distinct primes. If $J \subseteq \{1, \dots, t\}$ is nonempty, then*

$$1 + \sum_{j \in J} p_j \in \mathcal{R}.$$

Proof. Put $n = \prod_{j \in J} p_j$. Every composite divisor of n is larger than M^2 , whereas every prime divisor of n is at most $Y < M^2$. Consequently, the positive divisors of n begin with 1, followed by the primes p_j , $j \in J$, in increasing order. Their initial sum is the displayed integer. $\square$

Adjoining a prime extends an interval of subset sums when its translate meets or adjoins the original interval.

Lemma 3.4. *Let $C \leq U$ be integers. Suppose that the subset sums of a finite set of distinct primes contain every integer in $[C, U]$, and let p be a new prime satisfying $p \leq U - C + 1$. After adjoining p , the subset sums contain every integer in $[C, U + p]$.*

Proof. The old subset sums cover $[C, U]$, and the subset sums using p cover $[C + p, U + p]$. The hypothesis gives $C + p \leq U + 1$, so these two integer intervals overlap or are consecutive. $\square$

The finite checks and interval-extension argument cover the following three overlapping ranges.

Proposition 3.5. *Every positive integer in each of the following intervals is represented:*

$$[6, 10\,000\,000], \tag{3.1} \text{eq:fraiture-small}$$

$$[469\,616, 273\,803\,744\,799\,154], \tag{3.2} \text{eq:fraiture-first}$$

$$[105\,000\,001, 10^{27} + 10^8]. \tag{3.3} \text{eq:fraiture-large}$$

Proof. The three finite integer checks used in this proof are implemented in our verifier `verify.rs` at [9]. Its committed `output.json` records a successful run, together with interval endpoints and sample witnesses. The verifier constructs the witnesses as it runs; the JSON file does not contain all 51 000 001 large-seed witnesses.

For (3.1), take $1 \leq B \leq 1000$ and a prime $q > B$. The divisors of Bq consist first of the divisors of B and then of q times those divisors, in the same order. Thus every integer of the form

$$\sigma(B) + q \sum_{i=1}^j d_i(B), \quad 1 \leq j \leq \tau(B),$$

is represented. A sieve through 10 000 000 marks every integer in (3.1) except 7, which has the separate representation $7 = 1 + 2 + 4$ from $n = 4$.

For (3.2), take $M = 10\,000$ and $Y = 99\,000\,000 < M^2$. The first 94 primes in $(M, Y]$, lying between 10 007 and 10 883, have subset sums covering $[469\,615, 480\,503]$. This is checked by the exact recurrence $S_0 = \{0\}$, $S_j = S_{j-1} \cup (S_{j-1} + p_j)$ for $1 \leq j \leq 94$. The verifier checks the hypothesis of Lemma 3.4 for each successive prime through 98 999 987. The resulting subset-sum interval is

$$[469\,615, 273\,803\,744\,799\,153].$$

Lemma 3.3 then gives (3.2) after adding 1.

For (3.3), let $M = 20\,000\,000$. For every one of the 51 000 001 integers in

$$[105\,000\,000, 156\,000\,000],$$

the verifier constructs and checks a sum of four or five distinct primes in $(M, 2M)$. For completeness, the witness search is deterministic. For an odd target, first remove the fixed prime 20 000 003; for an even target, remove nothing. Split the resulting even number into even summands $e_1 + e_2$, beginning with the largest even e_1 not exceeding half that number and decreasing e_1 by 2, subject to $42\,000\,000 \leq e_i \leq 78\,000\,000$. For each e_i , search in increasing order for a prime p such that $p < e_i - p$, both p and $e_i - p$ lie in $(M, 2M)$, and neither has already been used. The verifier checks that this search succeeds for every target, and then checks each completed witness directly. The next prime after $2M$ is 40 000 003, which is at most the inclusive width 51 000 001 of the certified interval. Thus Lemma 3.4 starts the interval-extension induction. After a prime p has been adjoined, the new interval width is at least $2p$, while Bertrand's postulate supplies the next prime below $2p$; the induction therefore continues for every subsequent prime.

Set $Y = 399\,000\,000\,000\,000 < M^2$. It remains to show that

$$\sum_{2M < p \leq Y} p > 10^{27} + 10^8.$$

A single block suffices. Put $a = Y/2 = 199\,500\,000\,000\,000$. Dusart's explicit estimates [5, Theorem 6.9] give

$$\frac{x}{\log x} < \pi(x) \leq \frac{x}{\log x} \left(1 + \frac{1.2762}{\log x}\right) < 1.05 \frac{x}{\log x} \quad (x \geq a),$$

because $\log a > 32$. Since $\log Y < 34$ and $2M < a$, it follows that

$$\begin{aligned} \sum_{2M < p \leq Y} p &\geq a(\pi(Y) - \pi(a)) \\ &> a \left(\frac{Y}{34} - \frac{1.05a}{32} \right) \\ &= \frac{17\,599\,173\,046\,875\,000\,000\,000\,000\,000}{17} > 1.03 \cdot 10^{27} > 10^{27} + 10^8. \end{aligned}$$

The coarse logarithm inequalities require only $2.71 < e < 2.72$ and the rational comparisons $2.72^{32} < a$ and $Y < 2.71^{34}$. The final bound is an inequality between rational numbers, so this bridge does not depend on floating-point logarithms or on a rounding convention in the verifier. The extended subset-sum interval therefore reaches past $10^{27} + 10^8$; adding 1 and applying Lemma 3.3 proves (3.3). $\square$

For reproducibility, we record the commands and expected output of our finite verifier. Use the following verifier [9]. Run

```
rustc -O -C overflow-checks=on verify.rs -o /tmp/v
/tmp/v all
```

The final JSON object should have "check": "all" and "ok": true. The integer checks use a sieve and a subset-sum recurrence to verify interval overlap and the primality, distinctness, location, and sum of every large-seed witness. Specifically, the first check marks 9 999 994 of the 9 999 995 integers in $[6, 10^7]$, with 7 handled separately; the first prime window contains 5 705 894 primes; and the large-seed check examines 51 000 001 targets. The verifier's optional numerical bridge and tail checks are not needed for the rational and analytic proofs given here.

3.2. An explicit balanced ternary Goldbach problem.

Lemma 3.6. *Every odd integer $H \geq 10^{27}$ is the sum of three distinct odd primes p, q, r satisfying*

$$p, q, r > \frac{H}{30\,000 \log H}.$$

Proof. Helfgott's explicit argument [11, §7.4, equations (7.49)–(7.50)] gives a real weighted sum

$$\sum_{\substack{p+q+r=H \\ p,q,r \text{ odd primes}}} (\log p)(\log q)(\log r) \eta_+(p/x) \eta_+(q/x) \eta_*(r/x) \geq 0.000422 H^2,$$

where $x = H/(2 + 9/(196\sqrt{2\pi}))$, for every odd $H \geq 10^{27}$. Here η_+ and η_* are the real smoothing functions used in that proof. Their individual smoothing factors satisfy

$$\|\eta_+\|_\infty \leq 1.079955, \quad \|\eta_*\|_\infty \leq 1.414;$$

see [11, equations (7.3) and (7.19)]. Put

$$W = (1.079955)^2 \cdot 1.414, \quad z = \frac{H}{30\,000 \log H}.$$

The explicit estimate $\psi(t) \leq 1.03883t$ of [16] bounds the total absolute weight of triples with at least one coordinate at most z by

$$3W\psi(z)\psi(H)\log H \leq \frac{3W(1.03883)^2}{30\,000}H^2 < 0.000178H^2.$$

Triples with repeated coordinates have total absolute weight at most

$$3WH(\log H)^3 < 10^{-20}H^2 \quad (H \geq 10^{27}).$$

For the last inequality, $(\log H)^3/H$ decreases on this range, and $\log(10^{27}) < 63$. Since $0.000422 > 0.000178 + 10^{-20}$, some odd-prime triple remains after both exceptional classes have been discarded. $\square$

The balanced prime representation now proves representability throughout the remaining infinite range.

rop:fraiture-tail) **Proposition 3.7.** *Every integer*

$$n \geq 10^{27} + 10^8$$

belongs to $\mathcal{R}$.

Proof. First let n be even and put $H = n - 1$. Apply Lemma 3.6, and order its primes as $p < q < r$. Since $H/(\log H)^2$ increases for $H > e^2$ and

$$\frac{10^{27}}{(30\,000 \log(10^{27}))^2} > 1,$$

we have

$$pq > \left(\frac{H}{30\,000 \log H} \right)^2 > H > r.$$

Hence the divisors of pqr begin with $1, p, q, r$, and their sum is $1 + p + q + r = n$.

Now let n be odd. By Bertrand's postulate, choose an odd prime ℓ such that

$$60\,000 \log n < \ell < 120\,000 \log n,$$

and put $H = n - 1 - \ell$. Set $T = 10^{27} + 10^8$. The function

$$h(t) = t - 1 - 120\,000 \log t - 10^{27}$$

has positive derivative for $t \geq T$, and $h(T) > 0$ because $\log T < 64$. Therefore $H \geq 10^{27}$; in the same way, $H > n/2$. Apply Lemma 3.6 and order the resulting primes as $p < q < r$. Since $\log H < \log n$,

$$p > \frac{H}{30\,000 \log H} > \frac{n}{60\,000 \log n} > 120\,000 \log n > \ell.$$

For the penultimate inequality, observe that $n/(\log n)^2$ increases for $n > e^2$ and already exceeds $7.2 \cdot 10^9$ at $n = T$. Finally,

$$\ell p > 60\,000 \log n \cdot \frac{H}{30\,000 \log H} > 2H > r.$$

Thus the divisors of ℓpqr begin with $1, \ell, p, q, r$, whose sum is $1 + \ell + p + q + r = n$. $\square$

Proof of Theorem 3.2. The three intervals of Proposition 3.5 overlap, and the last one overlaps the infinite interval of Proposition 3.7 at $10^{27} + 10^8$. Hence every $n \geq 6$ is represented. The integers 1, 3, 4 are represented by $m = 1, 2, 3$, respectively. No prefix can equal 2: its first term is 1, and every longer prefix has sum at least $1 + 2 = 3$. Similarly, a two-term prefix equal to 5 would have to be $1 + 4$, but $4 \mid m$ implies $2 \mid m$, contradicting that 4 is the second-smallest divisor. Every prefix of length at least three has sum at least $1 + 2 + 3 = 6$. Therefore neither 2 nor 5 is represented. $\square$

The computational dependencies of the representability argument are as follows.

Remark 3.8. The exact exceptional set in Theorem 3.2 uses the three finite integer checks in Proposition 3.5. The infinite-tail Proposition 3.7 and the analytic Lemma 3.1 use none of these checks. In particular, the analytic proofs of the large-value estimates may use the latter lemma whenever representedness of odd targets is needed.

4. SMALL VALUES OF THE LEAST REPRESENTING INTEGER

4.1. Upper bounds for the lower tail. We extend the first-moment argument from Tao's comment on [2] to obtain the following higher-moment estimate.

(lem:kovac-moment) **Lemma 4.1.** *There is an absolute constant $C > 0$ such that, for every integer $q \geq 3$ and every real $x \geq 1$,*

$$\sum_{n \leq x} \sum_{j \geq 0} \left(\frac{\sigma_j(n)}{n} \right)^q \ll \exp(Cq \log \log q) x.$$

The implicit constant is absolute.

Proof. Let

$$1 = r_1 < r_2 < \cdots < r_{\tau(n)} = n$$

be the divisors of n . Divisor reflection gives

$$\frac{\sigma_j(n)}{n} = \sum_{i > j} \frac{1}{r_i}.$$

After expanding the q th power and summing over j , a fixed tuple $a_1, \dots, a_q$ of divisors is counted once for each divisor threshold not exceeding $\min_i a_i$. Hence

$$\sum_{j \geq 0} \left(\frac{\sigma_j(n)}{n} \right)^q = \sum_{\substack{r, a_1, \dots, a_q | n \\ r \leq \min(a_1, \dots, a_q)}} \frac{1}{a_1 \cdots a_q}.$$

Summing over $n \leq x$ and bounding the number of multiples of $\text{lcm}(r, a_1, \dots, a_q)$ by $x / \text{lcm}(r, a_1, \dots, a_q)$, it is enough to prove

$$S := \sum_{\substack{r, a_1, \dots, a_q \geq 1 \\ r \leq \min_i a_i}} \frac{1}{a_1 \cdots a_q \text{lcm}(r, a_1, \dots, a_q)} \ll \exp(Cq \log \log q). \quad (4.1) \quad \boxed{\text{eq:k-S}}$$

For each i , put

$$b_i = \frac{r}{\gcd(r, a_i)}, \quad c_i = \frac{a_i}{\gcd(r, a_i)}.$$

Then $b_i \mid r$, $\gcd(b_i, c_i) = 1$, $a_i = rc_i/b_i$, and

$$\text{lcm}(r, a_1, \dots, a_q) = r \text{lcm}(c_1, \dots, c_q).$$

The condition $r \leq a_i$ becomes $b_i \leq c_i$. Dropping the coprimality conditions and taking $\beta = (q-1)/q$, the inequality

$$\frac{1}{c_i} \leq \frac{1}{b_i^{1-\beta} c_i^\beta} \quad (c_i \geq b_i)$$

applies term by term. Replacing each divisor $b_i \mid r$ by r/d_i and using $\beta q = q-1$ leaves the power r^{-2} . Thus $S \leq S' S''$, where

$$S' = \sum_{r \geq 1} \frac{1}{r^2} \left(\sum_{d \mid r} \frac{1}{d^{(q-1)/q}} \right)^q$$

and

$$S'' = \sum_{c_1, \dots, c_q \geq 1} \frac{1}{c_1^{(q-1)/q} \dots c_q^{(q-1)/q} \text{lcm}(c_1, \dots, c_q)}.$$

Both sums have Euler products.

For S' , write $S' = \prod_p L_p$, where

$$L_p = \sum_{\ell \geq 0} p^{-2\ell} \left(\sum_{0 \leq \gamma \leq \ell} p^{-((q-1)/q)\gamma} \right)^q.$$

For $p \leq q$,

$$L_p \leq (1 - p^{-2})^{-1} (1 - p^{-(q-1)/q})^{-q}.$$

Since

$$\sum_{p \leq q} p^{-(q-1)/q} \leq q^{1/q} \sum_{p \leq q} \frac{1}{p} \ll \log \log q,$$

one obtains

$$\prod_{p \leq q} L_p \ll \exp(Cq \log \log q).$$

For $p > q$, one has $qp^{-(q-1)/q} \ll 1$, and the terms with $\ell \geq 1$ give $L_p = 1 + O(p^{-2})$. Thus

$$S' \ll \exp(Cq \log \log q). \quad (4.2) \quad \boxed{\text{eq:k-Sprime}}$$

For S'' , write $S'' = \prod_p M_p$, where

$$M_p = \sum_{\gamma_1, \dots, \gamma_q \geq 0} p^{-((q-1)/q)(\gamma_1 + \dots + \gamma_q) - \max_i \gamma_i}.$$

Put

$$A_h = \sum_{0 \leq \gamma \leq h} p^{-((q-1)/q)\gamma}.$$

Grouping by $h = \max_i \gamma_i$ gives

$$M_p = \sum_{h \geq 0} p^{-h} (A_h^q - A_{h-1}^q), \quad A_{-1} = 0.$$

For $p \leq q$,

$$M_p \leq (1 - p^{-1})^{-1} (1 - p^{-(q-1)/q})^{-q},$$

so the product over these primes is $\ll \exp(Cq \log \log q)$. For $p > q$, the mean-value theorem and $qp^{-(q-1)/q} \ll 1$ give

$$A_h^q - A_{h-1}^q \ll qp^{-h(q-1)/q}.$$

Consequently,

$$M_p - 1 \ll qp^{-2+1/q}, \quad \sum_{p > q} (M_p - 1) \ll q \sum_{n > q} n^{-2+1/q} \ll 1.$$

Hence

$$S'' \ll \exp(Cq \log \log q). \quad (4.3) \quad \boxed{\text{eq:k-Ssecond}}$$

Combining (4.2) and (4.3) proves (4.1) and the lemma. $\square$

Proof of Theorem 1.1. Let

$$\mathcal{E}_\delta(X) = \{N \leq X : N \in \mathcal{R}, f(N) \leq \delta N\}.$$

If $\delta X < 1$, then $\mathcal{E}_\delta(X)$ is empty, so assume $\delta X \geq 1$. If $N \in \mathcal{E}_\delta(X)$ and $n = f(N)$, then $n \leq \delta X$ and $N = \sigma_j(n)$ for some $j \geq 0$ with $\sigma_j(n)/n \geq 1/\delta$. Therefore, for every integer $q \geq 3$,

$$\begin{aligned} \#\mathcal{E}_\delta(X) &\leq \delta^q \sum_{n \leq \delta X} \sum_{j \geq 0} \left(\frac{\sigma_j(n)}{n} \right)^q \\ &\ll \delta^{q+1} \exp(Cq \log \log q) X. \end{aligned}$$

Choose a sufficiently small absolute $c > 0$ and put

$$q = \lfloor \exp((1/\delta)^c) \rfloor.$$

For all sufficiently small δ , one has $C \log \log q \leq \frac{1}{2} \log(1/\delta)$, and hence

$$\delta^{q+1} \exp(Cq \log \log q) \leq \exp\left(-\frac{q}{2} \log \frac{1}{\delta}\right) \leq \exp\left(-\exp((1/\delta)^{c/2})\right).$$

Renaming $c/2$ as c proves the first assertion. The fixed-power bound follows because this double-exponential expression is $O_M(\delta^M)$ for every fixed $M > 0$. $\square$

4.2. Where are the small ratios? The positive-density assertion for every fixed small ratio is compatible with the absence of such ratios at any practical search range. The following bound makes this distinction quantitative.

Proposition 4.2. *If $n \in \mathcal{R}$ and $f(n) \leq n/10$, then $n > 10^{120}$.*

Proof. Put $m = f(n)$. Every divisor-prefix sum of m is at most $\sigma(m)$, so $f(n) \leq n/10$ implies

$$\frac{\sigma(m)}{m} \geq \frac{n}{m} \geq 10.$$

If $m \leq 5040$, then

$$\frac{\sigma(m)}{m} = \sum_{d|m} \frac{1}{d} \leq \sum_{d=1}^{5040} \frac{1}{d} \leq 1 + \log 5040 < 10,$$

a contradiction. If $n \leq 10^{120}$, then $m \leq 10^{119}$; hence Axler's unconditional refinement of Robin's inequality [1, Corollary 2] gives

$$\frac{\sigma(m)}{m} < (1 + 3.15367 \cdot 10^{-7}) e^\gamma \log \log m \leq 9.99745 < 10,$$

again a contradiction. $\square$

The obstruction is not vacuous. For $m = \text{lcm}(1, \dots, 289)$, direct evaluation of the finite Euler product for $\sigma(m)/m$ gives

$$\frac{\sigma(m)}{m} = 10.004735741 \dots > 10.$$

Thus $n = \sigma(m)$ satisfies

$$n = 3.078152209 \dots \cdot 10^{128}, \quad \frac{f(n)}{n} \leq \frac{m}{\sigma(m)} = 0.099952665 \dots < \frac{1}{10}.$$

4.3. A positive-density family of abundant witnesses. We now prove Theorem 1.2 by adapting the positive-density range argument of Luca and Pomerance [12]. The main additional requirement is that the witnesses have arbitrarily large abundancy. We enforce it by a fixed divisor D . A theorem of Pollack [14] then bounds the abundancy of the intermediate aliquot sums; this replaces the deficiency condition used in [12]. After selecting a regular family, we partition it by its small-prime part and bound the number of pairs with the same aliquot sum. Cauchy-Schwarz will convert this collision bound into a positive-density image.

If $n \geq 2$, then $s(n)$ is the sum of all proper divisors of n , so $s(n) \in \mathcal{R}$ and

$$f(s(n)) \leq n \quad (n \geq 2). \tag{4.4} \quad \boxed{\text{eq:sv-basic}}$$

We will construct $\mathcal{A}(X) \subset [1, X]$ such that

$$n/\delta < s(n) \leq C_\delta X \quad (n \in \mathcal{A}(X)), \quad \#s(\mathcal{A}(X)) \gg_\delta X.$$

By (4.4), rescaling X then proves the theorem.

It is enough to prove the theorem for $0 < \delta \leq 1$. Put $\lambda = 1/\delta$, and choose a fixed squarefree integer D such that

$$\frac{\sigma(D)}{D} > 1 + \lambda. \quad (4.5) \quad \boxed{\text{eq:sv-D}}$$

Such a D exists because $\prod_{p \leq z} (1 + 1/p)$ is unbounded.

For large X , let $\mathcal{A}_0(X)$ consist of the integers

$$M = pqrk \quad (4.6) \quad \boxed{\text{eq:sv-factorizati}}$$

for which p, q, r are prime,

$$\begin{aligned} X^{1/120} < k = Dj &\leq X^{1/60}, \quad \frac{\sigma(j)}{j} \leq 4\zeta(2), \\ X^{1/15} < r &\leq X^{1/12}, \\ X^{7/20} < q &\leq X^{11/30}, \\ \frac{X}{2qrk} < p &\leq \frac{X}{qrk}. \end{aligned}$$

Set $\ell = rk$ and $n = q\ell$.

Lemma 4.3. *There is a constant $C_\delta > 0$ such that, for all sufficiently large X ,*

$$\#\mathcal{A}_0(X) \gg_\delta X,$$

every $M \in \mathcal{A}_0(X)$ has a unique factorization (4.6), and

$$\lambda < \frac{s(t)}{t} < C_\delta \quad (t \in \{k, \ell, n, M\}). \quad (4.7) \quad \boxed{\text{eq:sv-two-sided}}$$

Moreover, $q > n^{7/9}$.

Proof. For $J \geq 1$,

$$\sum_{j \leq J} \frac{\sigma(j)}{j^2} = \sum_{d \leq J} \frac{1}{d^2} \sum_{a \leq J/d} \frac{1}{a} \leq \zeta(2) \sum_{a \leq J} \frac{1}{a}.$$

Consequently, with $J_1 = X^{1/120}/D$ and $J_2 = X^{1/60}/D$,

$$\begin{aligned} \sum_{\substack{J_1 < j \leq J_2 \\ \sigma(j)/j \leq 4\zeta(2)}} \frac{1}{j} &\geq \sum_{J_1 < j \leq J_2} \frac{1}{j} - \frac{1}{4\zeta(2)} \sum_{j \leq J_2} \frac{\sigma(j)}{j^2} \\ &\geq \left(\frac{1}{240} + o(1) \right) \log X. \end{aligned}$$

For fixed q, r, k , the interval for p is $(T/2, T]$, where $T = X/(qrk) \geq X^{8/15}$. The prime number theorem, followed by Mertens' theorem for primes, therefore gives

$$\#\mathcal{A}_0(X) \gg \frac{X}{\log X} \sum_k \frac{1}{k} \sum_{X^{1/15} < r \leq X^{1/12}} \frac{1}{r} \sum_{X^{7/20} < q \leq X^{11/30}} \frac{1}{q} \gg_\delta X.$$

The factorization is unique because $p > X^{1/2}$, the prime q lies above $X^{7/20}$, the prime r lies between $X^{1/15}$ and $X^{1/12}$, and every prime factor of k is at most $X^{1/60}$.

The function $n \mapsto \sigma(n)/n$ is nondecreasing under divisibility. Since $D \mid t$ for $t \in \{k, \ell, n, M\}$, (4.5) therefore gives the lower bound in (4.7). The inequality $\sigma(ab)/(ab) \leq (\sigma(a)/a)(\sigma(b)/b)$ also gives

$$\frac{\sigma(k)}{k} \leq \frac{\sigma(D)}{D} \frac{\sigma(j)}{j} \leq 4\zeta(2) \frac{\sigma(D)}{D},$$

and adjoining each of p, q, r multiplies $\sigma(t)/t$ by less than 2. This gives the upper bound in (4.7). Finally, $\ell \leq X^{1/10}$ and $q^2 > X^{7/10} \geq \ell^7$, whence $q^9 > q^7 \ell^7 = n^7$. $\square$

4.4. Arithmetic regularity and smooth-part classes. For u sufficiently large, put

$$y(u) = \frac{\log \log u}{\log \log \log u}, \quad Y = y(X^{1/120}).$$

An integer is called Y -smooth if all of its prime factors are at most Y .

We collect the almost-everywhere regularity properties of divisor sums established by Luca and Pomerance.

(lem:LP-inputs)

Lemma 4.4. *Put $y(n) = \log \log n / \log \log \log n$. Outside a set of integers of asymptotic density zero, the following hold:*

- (i) $v_p(\sigma(n)) > v_p(n)$ for every prime $p \leq y(n)$;
- (ii) $P^+(\gcd(n, \sigma(n))) \leq y(n)$;
- (iii) $\sigma(n) / \gcd(n, \sigma(n))$ is divisible by every prime $p \leq y(n)$;
- (iv) every prime factor of $s(n) / \gcd(n, \sigma(n))$ exceeds $y(n)$.

If in addition $P^+(n) > n^{7/9}$, then, outside another density-zero set,

$$y(n) < q_0 \leq n^{10/27}, \quad q_0 \text{ prime} \implies q_0^2 \nmid s(n).$$

Finally, outside a set of asymptotic density zero,

$$\sum_{\substack{a \mid \sigma(n) \\ a > (\log \log n)^2}} \frac{1}{a} \leq 1.$$

Proof. The first four properties are the conclusions of [12, Lemma 2.1], with the valuation inequality in (i) stated explicitly in the proof of its part (4). The square-divisibility assertion is the consequence of [12, Lemma 2.2] needed here; we use it only in the displayed range. The reciprocal-divisor estimate is [12, Lemma 2.5]. $\square$

We next count integers with a prescribed smooth part and bound the proportion whose smooth part is large.

(smooth-part-input)

Lemma 4.5. *Suppose that $Y = Y(X) \rightarrow \infty$ and $Y = o(\log X)$. For every fixed $c > 0$, uniformly for Y -smooth $d \leq Y^c$,*

$$\#\{n \leq X : d \text{ is the largest } Y\text{-smooth divisor of } n\} \asymp \frac{X}{d \log Y}.$$

Moreover, given $\eta > 0$, c may be chosen sufficiently large that the integers $n \leq X$ whose largest Y -smooth divisor exceeds Y^c account for at most ηX integers for all sufficiently large X .

Proof. Write $D_Y(n)$ for the largest Y -smooth divisor of n . For a Y -smooth integer d , $D_Y(n) = d$ if and only if $n = du$ with $\gcd(u, Y\#) = 1$. Counting complete periods modulo $Y\#$ gives

$$\#\{n \leq X : D_Y(n) = d\} = \frac{X}{d} \Delta(Y) + O(Y\#).$$

Chebyshev's estimate gives $Y\# = \exp(O(Y)) = X^{o(1)}$, while Mertens' theorem gives $\Delta(Y) \asymp 1/\log Y$. This proves the required uniform estimate. It is the small- Y form of the counting argument in [12, (3.1)].

For the tail, sum the logarithm of the smooth part:

$$\sum_{n \leq X} \log D_Y(n) = \sum_{p \leq Y} \log p \sum_{a \geq 1} \left\lfloor \frac{X}{p^a} \right\rfloor \leq X \sum_{p \leq Y} \frac{\log p}{p-1} \ll X \log Y.$$

The last estimate follows by partial summation from Chebyshev's bound. Markov's inequality now gives

$$\#\{n \leq X : D_Y(n) > Y^c\} \ll X/c.$$

Choosing c sufficiently large in terms of η proves the assertion. $\square$

The following lemma selects a large subfamily with the regularity properties needed in the collision estimate.

`<lem:sv-regular>`

Lemma 4.6. *There is a subfamily $\mathcal{A}(X) \subseteq \mathcal{A}_0(X)$ with $\#\mathcal{A}(X) \gg_\delta X$ having the following properties. For $M = pqrk \in \mathcal{A}(X)$, let d be the largest Y -smooth divisor of M . Then, for every $t \in \{k, \ell, n, M\}$,*

- (i) $d = \gcd(t, \sigma(t))$, and d is the largest Y -smooth divisor of both t and $s(t)$;
- (ii) $\sigma(t)/d$ is divisible by every prime at most Y , while every prime factor of $s(t)/d$ exceeds $y(t)$.

In addition,

$$y(n) < q_0 \leq n^{10/27}, \quad q_0 \text{ prime} \implies q_0^2 \nmid s(n),$$

and

$$\sum_{\substack{a|\sigma(n) \\ a > (\log \log X)^2}} \frac{1}{a} \leq 1. \tag{4.8} \text{eq:sv-LP25}$$

There is also a constant $B_\delta > 0$ such that

$$\frac{\sigma(s(n))}{s(n)} \leq B_\delta. \tag{4.9} \text{eq:sv-image-abund}$$

Proof. First discard the integers for which k has a prime factor in $(Y, y(X)]$. Uniformly for $X^{1/120} < k \leq X^{1/60}$,

$$y(k) = y(X) + O\left(\frac{1}{\log \log \log X}\right),$$

so

$$\sum_{Y < p \leq y(X)} \frac{1}{p} = o(1).$$

Indeed,

$$\sum_{\substack{k \leq X^{1/60} \\ p|k \text{ for some } Y < p \leq y(X)}} \frac{1}{k} \ll \log X \sum_{Y < p \leq y(X)} \frac{1}{p} = o(\log X).$$

The discarded k therefore account for $o(X)$ members of $\mathcal{A}_0(X)$.

Apply Lemma 4.4 to each of k, ℓ, n, M , with its large-prime square-divisibility conclusion applied to n .

Each excluded set has asymptotic density zero. These exclusions remove only $o(X)$ members of $\mathcal{A}_0(X)$. To justify this assertion, let $\mathcal{E} \subseteq \mathbb{N}$ satisfy $\#(\mathcal{E} \cap [1, T]) = o(T)$. Partial summation gives

$$\sum_{\substack{n \leq T \\ n \in \mathcal{E}}} \frac{1}{n} = o(\log T). \tag{4.10} \text{eq:sv-harmonic-de}$$

The separated factor ranges make the decompositions $\ell = rk$, $n = q\ell$ and $M = pn$ unique. An upper-bound estimate for the prime p then shows that the members for which one of k, ℓ, n lies in $\mathcal{E}$ are

$$\ll \frac{X}{\log X} \sum_{\substack{t \leq X^{O(1)} \\ t \in \mathcal{E}}} \frac{1}{t} = o(X),$$

up to bounded reciprocal prime sums; the deletion at the level M is directly $o(X)$.

For every retained t , Lemma 4.4 and its first four properties identify $\gcd(t, \sigma(t))$ with the largest $y(t)$ -smooth divisor of t and give the corresponding divisibility properties of $\sigma(t)$ and $s(t)$. Since $Y \leq y(t) \leq y(X)$, the exclusion made in the first paragraph and the fact that $p, q, r > y(X)$ show that the same integer d occurs at all four levels and prove (i)–(ii). The hypothesis $q > n^{7/9}$ gives the displayed square-divisibility exclusion for $s(n)$, and the final assertion of Lemma 4.4 gives (4.8).

By [14, Theorem 1.4], the fixed exceptional set of integers n for which $s(s(n))/s(n) > s(n)/n + 1$ has asymptotic density zero. One further exclusion at the level n , justified by (4.10), therefore ensures that

$$\frac{s(s(n))}{s(n)} \leq \frac{s(n)}{n} + 1.$$

Together with (4.7), this yields (4.9) with $B_\delta = C_\delta + 2$. □

Partition $\mathcal{A}(X)$ according to the integer d in Lemma 4.6. The image sets belonging to distinct classes are disjoint, because d is also the largest Y -smooth divisor of $s(M)$.

lem:sv-classes **Lemma 4.7.** *There are positive constants c, c_1, c_2 , depending at most on δ , and a set $\mathcal{D}_X$ of Y -smooth integers $d \leq Y^c$ such that, writing*

$$\mathcal{A}_d(X) = \{M \in \mathcal{A}(X) : d \text{ is the largest } Y\text{-smooth divisor of } M\},$$

one has, uniformly for $d \in \mathcal{D}_X$,

$$c_1 \frac{X}{d \log Y} \leq \#\mathcal{A}_d(X) \leq c_2 \frac{X}{d \log Y}, \quad \sum_{d \in \mathcal{D}_X} \frac{1}{d} \gg_\delta \log Y. \quad (4.11) \quad \boxed{\text{eq:sv-class-size}}$$

The sets $s(\mathcal{A}_d(X))$, $d \in \mathcal{D}_X$, are pairwise disjoint.

Proof. Fix $\eta = \eta(\delta) > 0$ with $\#\mathcal{A}(X) \geq \eta X$. Apply Lemma 4.5, choosing c so large that the classes with $d > Y^c$ contain at most $\eta X/3$ integers. Since

$$\sum_{P^+(d) \leq Y} \frac{1}{d} = \prod_{p \leq Y} \left(1 - \frac{1}{p}\right)^{-1} \asymp \log Y,$$

we may choose a sufficiently small $a = a(\delta) > 0$ and let $\mathcal{D}_X$ consist of those $d \leq Y^c$ for which

$$\#\mathcal{A}_d(X) \geq a \frac{X}{d \log Y}.$$

The classes outside $\mathcal{D}_X$ then contain at most $\eta X/3$ integers. Thus the classes in $\mathcal{D}_X$ contain $\gg_\delta X$ integers. The uniform upper estimate above now gives

$$\frac{X}{\log Y} \sum_{d \in \mathcal{D}_X} \frac{1}{d} \gg_\delta X,$$

which proves (4.11). The disjointness assertion follows from Lemma 4.6. □

4.5. The collision estimate.

(sv-second-moment)

Proposition 4.8. *For $d \in \mathcal{D}_X$, put*

$$R_d(u) = \#\{M \in \mathcal{A}_d(X) : s(M) = u\}.$$

Then, uniformly for $d \in \mathcal{D}_X$,

$$\sum_u R_d(u)^2 \ll_\delta \frac{X}{d \log Y}. \quad (4.12) \quad \text{eq:sv-second-mome}$$

Proof. We prove the collision estimate directly, following the organization of [12, (3.4), §§3.1–3.2]. All estimates needed for the present family are recorded below. We first note two reciprocal estimates. Since d is the largest Y -smooth divisor of every member of $\mathcal{A}_d(X)$, the corresponding factor k is d times an integer with no prime factor at most Y . The complete-period count in Lemma 4.5, applied uniformly throughout the fixed power interval for k , and partial summation give

$$\sum_k \frac{1}{k} \ll \frac{\log X}{d \log Y}. \quad (4.13) \quad \text{eq:sv-k-reciproca}$$

Here and below a sum over a factor of a member of $\mathcal{A}_d(X)$ is restricted to the ranges in (4.6). The reciprocal prime sums over the fixed power intervals for q and r are bounded, so

$$\sum_n \frac{1}{n} \ll \frac{\log X}{d \log Y}. \quad (4.14) \quad \text{eq:sv-m-reciproca}$$

The diagonal contribution to $\sum_u R_d(u)^2$ is $\#\mathcal{A}_d(X)$ and is bounded by (4.11). Consider an off-diagonal collision, written as $M = pn$, $M' = p'n'$, with $n \neq n'$. Then

$$p s(n) + \sigma(n) = p' s(n') + \sigma(n'). \quad (4.15) \quad \text{eq:sv-collision}$$

By Lemma 4.6, $\gcd(n, \sigma(n)) = \gcd(n', \sigma(n')) = d$. Put

$$\gcd(s(n), s(n')) = dh.$$

Every prime factor of h exceeds both $y(n)$ and $y(n')$. Equation (4.15) shows that dh divides $\sigma(n) - \sigma(n')$. This difference is nonzero. Indeed, equality would give $ps(n) = p's(n')$; for all sufficiently large X ,

$$\min(p, p') > \frac{1}{2} X^{8/15} > C_\delta X^{7/15} \geq \max\{s(n), s(n')\},$$

which forces $p = p'$, then $s(n) = s(n')$, and finally $n = n'$. Thus

$$A_1 = \frac{s(n)}{dh}, \quad A_2 = \frac{s(n')}{dh}, \quad A_3 = \frac{|\sigma(n) - \sigma(n')|}{dh}$$

are positive integers, with $\gcd(A_1, A_2) = 1$.

For fixed n, n' , elementary linear Diophantine theory shows that there are fixed integers p_0, p'_0 for which every integral solution has the form

$$p = p_0 + A_2 t, \quad p' = p'_0 + A_1 t \quad (t \in \mathbb{Z}).$$

All relevant values of t lie in an interval of length at most

$$T := \frac{X dh}{n s(n')}.$$

The bounds

$$\lambda n' \leq s(n') \leq C_\delta n', \quad n n' \leq X^{14/15},$$

give $T \gg_\delta X^{1/15}$ and $T \ll_\delta X dh / (n n')$, so $\log T \asymp \log X$. The affine forms are primitive whenever a prime pair exists. For example, a common divisor of p_0 and A_2 would divide every value of $p = p_0 + A_2 t$, whereas $p > A_2$ for all sufficiently large X ; the second form is identical. For odd primes the two primitive forms exclude at most two residue classes, and the prime 2

either leaves an allowed parity class or there are no solutions. The standard two-dimensional upper-bound sieve, in the form used in [12, (3.7)], therefore gives

$$\ll_{\delta} \frac{Xdh}{nn'(\log X)^2} \frac{A_1}{\varphi(A_1)} \frac{A_2}{\varphi(A_2)} \frac{A_3}{\varphi(A_3)}$$

possible prime pairs.

If $v \mid s(t)$, then

$$\frac{v}{\varphi(v)} \leq \zeta(2) \frac{\sigma(v)}{v} \leq \zeta(2) \frac{\sigma(s(t))}{s(t)} \leq \zeta(2) B_{\delta} \quad (t \in \{n, n'\}). \quad (4.16) \quad \text{eq:sv-totient}$$

This bounds the factors belonging to A_1, A_2 , and also gives $\varphi(h) \gg_{\delta} h$.

We now reduce the factor belonging to A_3 . Let $A_{3,1}$ contain its prime factors at most $(\log \log X)^2$, let $A_{3,2}$ contain those in $((\log \log X)^2, \log X]$, and let $A_{3,3}$ contain those exceeding $\log X$. Since $A_3 \ll_{\delta} X^{7/15}$,

$$\frac{A_{3,1}}{\varphi(A_{3,1})} \ll \log Y, \quad \frac{A_{3,3}}{\varphi(A_{3,3})} \ll 1.$$

Let $A'_{3,2}$ be the largest divisor of $A_{3,2}$ coprime to $\sigma(n)$. The primes removed from $A_{3,2}$ divide $\sigma(n)$ and exceed $(\log \log X)^2$, so (4.8) gives

$$\frac{A_{3,2}}{\varphi(A_{3,2})} \ll \frac{A'_{3,2}}{\varphi(A'_{3,2})}.$$

Consequently, it remains to prove

$$\frac{X \log Y}{(\log X)^2} \sum_{n \neq n'} \frac{dh}{nn'} \frac{A'_{3,2}}{\varphi(A'_{3,2})} \ll_{\delta} \frac{X}{d \log Y}, \quad (4.17) \quad \text{eq:sv-reduced-col}$$

where the sum is restricted to pairs n, n' admitting an off-diagonal collision in (4.15), and $dh = \gcd(s(n), s(n'))$. In particular, $dh \mid \sigma(n) - \sigma(n')$. Notice also that A_3 is divisible by every prime at most Y . Indeed, both $\sigma(n)/d$ and $\sigma(n')/d$ have this property by Lemma 4.6, while every prime factor of h exceeds Y .

Suppose first that $h > X^{10/33}$. Write $n = q\ell$ and $n' = q'\ell'$. We claim that $\gcd(h, s(\ell)\sigma(\ell)) = 1$. If a prime q_0 divided both h and $\sigma(\ell)$, then the identity $s(n) = qs(\ell) + \sigma(\ell)$ would imply either $q_0 = q$ or $q_0 \mid s(\ell)$. In the first case, $q_0 \mid \gcd(n, \sigma(n)) = d$; in the second case, $q_0 \mid \ell$ and $q_0 \mid \sigma(\ell)$, so again $q_0 \mid d$. Both conclusions contradict the fact that every prime factor of h exceeds Y , whereas d is Y -smooth. If instead $q_0 \mid h$ and $q_0 \mid s(\ell)$, then the same identity gives $q_0 \mid \sigma(\ell)$, which has just been excluded. Thus both $s(\ell)$ and $\sigma(\ell)$ are units modulo h . It follows that q is fixed in a reduced residue class modulo h , and the same holds for q' . Moreover,

$$n \equiv \sigma(n) \equiv \sigma(n') \equiv n' \pmod{h}.$$

Eliminating q, q' gives

$$s(\ell')\ell\sigma(\ell) - s(\ell)\ell'\sigma(\ell') \equiv 0 \pmod{h}. \quad (4.18) \quad \text{eq:sv-large-h-con}$$

The absolute value of the left side is at most

$$2C_{\delta}(C_{\delta} + 1) \max(\ell, \ell')^3 \ll_{\delta} X^{3/10} = o(X^{10/33}),$$

so it vanishes. For every prime power $q_0^a \parallel d$, Lemma 4.6 gives $q_0^a \mid s(\ell)$ and $q_0^{a+1} \mid \sigma(\ell)$; it also excludes primes exceeding Y from $\gcd(\ell, s(\ell))$. Hence

$$\gcd(\ell^2, s(\ell)) = d, \quad \gcd((\ell')^2, s(\ell')) = d.$$

The equality in (4.18) can be written as

$$\frac{\ell^2}{s(\ell)} + \ell = \frac{(\ell')^2}{s(\ell')} + \ell'.$$

The reduced denominators of the two fractions are $s(\ell)/d$ and $s(\ell')/d$. Since their difference is an integer, these denominators are equal. Thus $s(\ell) = s(\ell')$, and the displayed equality, viewed as a strictly increasing quadratic in ℓ , gives $\ell = \ell'$.

In this range Mertens' theorem gives $A'_{3,2}/\varphi(A'_{3,2}) \ll \log \log X / \log Y$. Summing the left side of (4.17) and using $\ell = \ell'$ gives

$$\ll \frac{Xd \log \log X}{(\log X)^2} \sum_{q, q', \ell, h} \frac{h}{qq'\ell^2}.$$

We may assume $q > q'$ by symmetry. Because $\ell = \ell'$, the two primes belong to the same residue class modulo h . Thus $q = q' + jh$ with $j \geq 1$, and

$$\sum_q \frac{1}{q} \leq \sum_{1 \leq j \leq X} \frac{1}{jh} \ll \frac{\log X}{h},$$

even if primality is discarded. Summing next over $h \mid s(q'\ell)$ is bounded by $\tau(s(q'\ell)) = X^{o(1)}$. The reciprocal sum over q' is bounded and $\sum_\ell \ell^{-2} \ll X^{-1/15}$. The contribution is therefore

$$dX^{14/15+o(1)} = o\left(\frac{X}{d \log Y}\right),$$

uniformly for $d \leq Y^c$.

It remains to treat $h \leq X^{10/33}$. Let $A''_{3,2}$ be the largest divisor of $A'_{3,2}$ coprime to $s(n')$. Equation (4.16) gives

$$\frac{A'_{3,2}}{\varphi(A'_{3,2})} \ll_\delta \frac{A''_{3,2}}{\varphi(A''_{3,2})}.$$

We first note that h is squarefree in this range. Indeed, if $q_0^2 \mid h$, then $q_0^2 \mid s(n)$ and $q_0 > y(n)$. By Lemma 4.4, this forces $q_0 > n^{10/27}$. Since $n > X^{17/40}$,

$$q_0^2 > n^{20/27} > X^{17/54} > X^{10/33} \geq h,$$

a contradiction.

Fix n', h, k . The congruences

$$qrk \equiv (q+1)(r+1)\sigma(k) \equiv n' \pmod{h}$$

may therefore be solved for the required symmetric functions. Indeed, any prime dividing both h and either n or $\sigma(n)$ would divide $\gcd(n, \sigma(n)) = d$, which is impossible because every prime factor of h exceeds Y . Thus k and $\sigma(k)$ are units modulo h . The two congruences therefore determine qr and $q+r$ modulo h . Since h is squarefree, there are at most $\tau(h)$ possible ordered residue pairs

$$q \equiv a \pmod{h}, \quad r \equiv b \pmod{h}.$$

For one such pair put

$$\mathfrak{f}(qr) = \sum_{\substack{(\log \log X)^2 < q_0 \leq \log X \\ q_0 \mid \sigma(kqr) - \sigma(n') \\ q_0 \nmid h\sigma(kqr)}} \frac{1}{q_0},$$

where q_0 is prime. Mertens' theorem gives

$$\frac{A''_{3,2}}{\varphi(A''_{3,2})} \ll_\delta 1 + \frac{\log \log X}{\log Y} \mathfrak{f}(qr). \tag{4.19} \quad \boxed{\text{eq:sv-intermediat}}$$

Indeed, when $\mathfrak{f}(qr) \leq 1$, the logarithm of the left side is $O(\mathfrak{f}(qr))$. When $\mathfrak{f}(qr) > 1$, Mertens' theorem bounds the contribution of the full prime range by $O(\log \log X / \log Y)$.

For fixed n', h, k, r and a prime q_0 occurring in $\mathfrak{f}(qr)$, the congruence

$$(q+1)\sigma(kr) \equiv \sigma(n') \pmod{q_0}$$

fixes q modulo q_0 . The condition $q_0 \nmid \sigma(kqr)$ ensures that $\sigma(kr)$ is invertible modulo q_0 . Since $q_0 \nmid h$, this congruence combines with $q \equiv a \pmod{h}$. Brun–Titchmarsh and partial summation, followed by (4.16), give

$$\sum_{q,r} \frac{f(qr)}{qr} \ll_{\delta} \begin{cases} \frac{\log X}{hX^{1/20}}, & X^{1/20} < h \leq X^{10/33}, \\ \frac{1}{h^2(\log \log X)^2}, & h \leq X^{1/20}. \end{cases} \quad (4.20) \quad \boxed{\text{eq:sv-f-sum}}$$

Here the sums are over the fixed residue pair. To see (4.20), first note that a nonreduced residue class modulo q_0 contains no admissible q , because $q > X^{7/20} > \log X \geq q_0$. Otherwise the combined class modulo $q_0 h$ is reduced. Uniformly in the present ranges, $q_0 h \leq X^{10/33} \log X < X^{7/20-\epsilon_0}$ for a fixed $\epsilon_0 > 0$, so Brun–Titchmarsh and partial summation bound its reciprocal prime sum by $O(1/\varphi(q_0 h)) \ll_{\delta} 1/(q_0 h)$. Summing the extra weight $1/q_0$ gives $O(1/(h(\log \log X)^2))$. The reciprocal sum over r in its class is

$$\ll_{\delta} \begin{cases} (\log X)X^{-1/20}, & h > X^{1/20}, \\ h^{-1}, & h \leq X^{1/20}, \end{cases}$$

which proves the estimate. The same progression bounds without f and (4.19) yield

$$\sum_{q,r} \left(\frac{1}{qr} + \frac{\log \log X}{\log Y} \frac{f(qr)}{qr} \right) \ll_{\delta} \begin{cases} \frac{(\log X)(\log \log X)}{hX^{1/20} \log Y}, & X^{1/20} < h \leq X^{10/33}, \\ h^{-2}, & h \leq X^{1/20}. \end{cases} \quad (4.21) \quad \boxed{\text{eq:sv-qr-sum}}$$

Using (4.13), the contribution for fixed n', h to (4.17) is at most

$$\frac{X \log Y}{(\log X)^2} \frac{dh}{n'} \frac{\log X}{d \log Y} \tau(h) \times \text{the applicable bound in (4.21)}.$$

For $X^{1/20} < h \leq X^{10/33}$ this is

$$\ll_{\delta} \frac{X^{19/20} \log \log X}{\log Y} \frac{\tau(h)}{n'}.$$

Now $\sum_{h|s(n')} \tau(h) \leq \tau(s(n'))^2 = X^{o(1)}$, and (4.14) shows that the total is

$$\frac{X^{19/20+o(1)}}{d} = o\left(\frac{X}{d \log Y}\right).$$

For $h \leq X^{1/20}$ the fixed- n', h contribution is

$$\ll_{\delta} \frac{X}{\log X} \frac{\tau(h)}{hn'}.$$

Finally,

$$\sum_{h|s(n')} \frac{\tau(h)}{h} \leq \left(\frac{\sigma(s(n'))}{s(n')} \right)^2 \leq B_{\delta}^2.$$

Summing over h and then using (4.14) gives $O_{\delta}(X/(d \log Y))$. Together with the large- h and diagonal contributions, this proves (4.12). $\square$

Proof of Theorem 1.2. For each $d \in \mathcal{D}_X$, Cauchy–Schwarz, Lemma 4.7, and Proposition 4.8 give

$$\#s(\mathcal{A}_d(X)) \geq \frac{\#\mathcal{A}_d(X)^2}{\sum_u R_d(u)^2} \gg_{\delta} \frac{X}{d \log Y}.$$

The image sets are disjoint, and hence

$$\#s(\mathcal{A}(X)) \gg_{\delta} \frac{X}{\log Y} \sum_{d \in \mathcal{D}_X} \frac{1}{d} \gg_{\delta} X.$$

If $N = s(M)$ with $M \in \mathcal{A}(X)$, then (4.4) and (4.7) give

$$f(N) \leq M < \delta N.$$

Moreover, $N = s(M) < C_\delta M \leq C_\delta X$. Replacing X by X/C_δ proves the theorem for $0 < \delta \leq 1$, and the case $\delta > 1$ follows from $\delta = 1$. $\square$

5. UPPER TAILS

The moment bound handles representations with large cofactors. For small cofactors, we use targets that are coprime to all small primes. Divisibility of $\sigma(n)$ then forces the omitted-divisor kernel to be unusually large. The following estimate quantifies the scarcity of those kernels.

5.1. Divisor tails of a least common multiple.

(lem:kernel-tails) **Lemma 5.1.** *For $y, H \geq 2$, put*

$$S(y, H) := \sum_{\substack{L | \Lambda(y) \\ L > H}} \frac{\tau(L)}{L}.$$

If $E \rightarrow \infty$ and

$$P := \left\lfloor E^{1+(2+(\log \log \log E)^{-1/2}) \log \log E / \log \log \log E} \right\rfloor,$$

then

$$\log S(E, P/E) \leq -\frac{\log \log E}{\sqrt{\log \log \log E}} + O\left(\frac{\log \log E}{\log \log \log E}\right). \quad (5.1) \quad \text{eq:fixed-kernel-t}$$

If instead $J \rightarrow \infty$,

$$W = \sqrt{\frac{J \log J}{2}}, \quad F = \lfloor e^W \rfloor,$$

then, for every fixed $C > 0$,

$$S\left(F, \frac{e^J}{CW}\right) \leq \exp\{-W + o(W)\}. \quad (5.2) \quad \text{eq:moving-kernel-t}$$

Proof. For $0 < \alpha \leq 1/4$, Rankin's inequality and the Euler product for the divisor function give

$$S(y, H) \leq H^{-\alpha} \prod_{p \leq y} \sum_{a \geq 0} \frac{a+1}{p^{a(1-\alpha)}} = H^{-\alpha} \prod_{p \leq y} (1 - p^{-1+\alpha})^{-2}. \quad (5.3) \quad \text{eq:rankin-kernel-t}$$

Mertens' estimate for $\sum_{p \leq y} 1/p$, Chebyshev's bound, and partial summation show that, when $z = \alpha \log y \rightarrow \infty$,

$$\sum_{p \leq y} p^{-1+\alpha} = \log \log y + O\left(\int_0^z \frac{e^u - 1}{u} du\right) = \log \log y + O\left(\frac{e^z}{z}\right). \quad (5.4) \quad \text{eq:sharp-prime-su}$$

The higher terms in the logarithm of the Euler product in (5.3) are $O(1)$ uniformly for $\alpha \leq 1/4$.

For (5.1), take $\alpha = \log \log \log E / \log E$. Then

$$\alpha \log(P/E) = \left(2 + \frac{1}{\sqrt{\log \log \log E}}\right) \log \log E + o(1),$$

whereas (5.4) gives

$$2 \sum_{p \leq E} p^{-1+\alpha} = 2 \log \log E + O\left(\frac{\log \log E}{\log \log \log E}\right).$$

Substitution in (5.3) proves the first assertion.

For (5.2), take $\alpha = W/J$. Since

$$\alpha \log F = \frac{1}{2} \log J + o(1), \quad \alpha \log \left(\frac{e^J}{CW} \right) = W + o(W),$$

equation (5.4) gives

$$\sum_{p \leq F} p^{-1+\alpha} \ll \log W + \frac{\sqrt{J}}{\log J} = o(W).$$

The second assertion follows once more from (5.3). $\square$

5.2. Periodic obstructions for bounded cofactors. For sufficiently large A , abbreviate the upper-tail scale by

$$L(A) := \frac{\log \log \log A}{(\log A)(\log \log A)}.$$

The forced moduli identify a periodic set that almost completely avoids the bounded-cofactor range. Its density estimate will also be useful for the coverage questions in Section 6.

For $A \geq 2$ let $\mathcal{K}_A$ be the set of moduli $K_{e,d}$ with $2 \leq e \leq A$ and $d \geq 1$; it is finite by Section 2. Put

$$V_A = \{N \in \mathbb{N} : k \nmid N \text{ for every } k \in \mathcal{K}_A\}.$$

This is a union of residue classes modulo $W_A = \text{lcm}(\mathcal{K}_A)$, so it has a natural density, and $d(V_A) \geq \varphi(W_A)/W_A > 0$. By inclusion and exclusion the density is the rational number

$$d(V_A) = 1 - \sum_{\emptyset \neq I \subseteq \mathcal{K}_A} \frac{(-1)^{|I|+1}}{\text{lcm}(I)}.$$

Proposition 5.2. *Fix $A \geq 2$. All but $o_A(X)$ of the integers $N \leq X$ in G_A are divisible by some $k \in \mathcal{K}_A$. Consequently*

$$1 - \underline{d}(G_A) \geq d(V_A).$$

Proof. Let $N \leq X$ lie in $G_A \cap V_A$. If N is a value of F_1 only, then $N = \sigma(d)$ lies in $\sigma(\mathbb{N})$, which has density zero by Lemma 2.4. Otherwise $N = F_e(d)$ with $2 \leq e \leq A$. Put $n = ed \leq AX$ and $k = K_{e,d} \in \mathcal{K}_A$. By (2.3), $N \equiv \sigma(n) \pmod{k}$, and $k \nmid N$ forces $k \nmid \sigma(n)$, hence $W_A \nmid \sigma(n)$. The pair (e, n) determines N , so these N number at most

$$A \cdot \#\{n \leq AX : W_A \nmid \sigma(n)\} = o_A(X)$$

by Lemma 2.3. Thus all but $o_A(X)$ of the elements of $V_A \cap [1, X]$ lie outside G_A , which gives the density bound. $\square$

The next estimate determines the scale supplied by this periodic obstruction.

Corollary 5.3. *As $A \rightarrow \infty$,*

$$\left(\frac{e^{-\gamma}}{2} - o(1) \right) L(A) \leq d(V_A) \leq \Delta(A).$$

Proof. Every prime $p \leq A$ lies in $\mathcal{K}_A$: with $e = p$ and $d = 1$ one has $D = \{1\}$, $M = C = g = 1$ and $K_{p,1} = p$. So V_A consists of A -rough integers, and $d(V_A) \leq \Delta(A)$.

For the lower bound let P be the parameter of Lemma 5.1 with $E = A$. Then $\log P = (2 + o(1))(\log A)(\log \log A)/\log \log \log A$, and Mertens' theorem gives

$$\Delta(P) = \left(\frac{e^{-\gamma}}{2} + o(1) \right) L(A).$$

It suffices to show that all but a proportion $o(1)$ of the P -rough integers lie in V_A .

Let N be P -rough and divisible by $k = K_{e,d}$ for some $2 \leq e \leq A$. Then k is P -rough. Since e/g divides k and $e/g \leq A < P$, we must have $e/g = 1$, that is, $e \mid M$ and $k = C$. Now $C \geq M \geq e \geq 2$, so $C > P$, and $C \leq (e-1)M$ gives $M > P/A$.

Given M , there are at most $\tau(M)$ choices of e , since $e \mid M$, and e and M together determine $D = \{j < e : j \mid M\}$ and hence C . So at most $\tau(M)$ moduli k share a given M , and for each of them the P -rough multiples of C have density $\Delta(P)/C \leq \Delta(P)/M$.

Since $M \mid \Lambda(A)$ and $M > P/A$,

$$d\{N : N \text{ } P\text{-rough, } N \notin V_A\} \leq \Delta(P) \sum_{\substack{M \mid \Lambda(A) \\ M > P/A}} \frac{\tau(M)}{M} = \Delta(P) S(A, P/A) = o(\Delta(P))$$

by (5.1). Hence $d(V_A) \geq (1 - o(1))\Delta(P)$. $\square$

5.3. Fixed upper levels.

Proof of Theorem 1.3. Fix an integer $j \geq 3$ and, for sufficiently large T , put

$$b_j = \frac{j+1}{j-1}, \quad E = \left\lceil T^{b_j} (\log T)^{3/(j-1)} \right\rceil.$$

Throughout the counting argument, j, T, E are fixed before $X \rightarrow \infty$. By Proposition 5.2, all but $o_E(X)$ of the targets in $V_E \cap [1, X]$ have no representation with cofactor at most E . Lemma 2.6 bounds the targets with a representation $n \leq TN$ and cofactor greater than E by

$$C_j T^{j+1} E^{1-j} X \leq \frac{C_j X}{(\log T)^3}.$$

Moreover, V_E consists of E -rough integers. Its nonsquarefree members up to X therefore number at most

$$X \sum_{p > E} \frac{1}{p^2} \ll \frac{X}{E}.$$

All its members are odd, so Lemma 3.1 discards only $o(X/\log \log \log X)$ unrepresented targets. Every target surviving these deletions is represented, squarefree, and satisfies $f(N) > TN$. Consequently

$$\begin{aligned} d\{N \in \mathcal{R} : N \text{ squarefree, } f(N) > TN\} \\ \geq d(V_E) - \frac{C_j}{(\log T)^3} - O(E^{-1}). \end{aligned}$$

Corollary 5.3 and

$$L(E) = \left(\frac{1}{b_j} + o(1) \right) L(T) \quad (T \rightarrow \infty)$$

show that the right-hand side is at least

$$\left(\frac{e^{-\gamma}}{2b_j} - o(1) \right) \frac{\log \log \log T}{(\log T)(\log \log T)}.$$

Here both error terms are $o(L(T))$ for each fixed j . First let $T \rightarrow \infty$ for fixed j , and then let $j \rightarrow \infty$. Since $b_j \rightarrow 1$, this proves (1.2), independently of the finite representability computation. $\square$

5.4. Growing upper levels.

Proof of Theorem 1.4. Fix $0 < \varepsilon < 1$ sufficiently small in terms of η . Choose a fixed $\rho > 0$ sufficiently small, depending only on the absolute constants in Lemma 2.5, and put

$$P = \left\lfloor \rho \frac{\log \log X}{\log \log \log X} \right\rfloor, \quad J = \log P, \quad W = \sqrt{\frac{J \log J}{2}}, \quad F = \lfloor e^W \rfloor, \quad t = e^{(1-\varepsilon)W}.$$

In particular $F < P$ for all sufficiently large X . Complete periods modulo $P\#$ and Mertens' theorem give

$$\#\{X/2 < N \leq X : \gcd(N, P\#) = 1\} = \frac{X}{2} \Delta(P) + O(P\#) \asymp \frac{X}{\log \log \log X}, \quad (5.5) \quad \boxed{\text{eq:sharp-rough-ta}}$$

because $P\# = \exp(O(P)) = X^{o(1)}$ and $\Delta(P) \asymp 1/J$. The nonsquarefree members of this interval number at most

$$X \sum_{p>P} \frac{1}{p^2} \ll \frac{X}{P} = o(X\Delta(P)).$$

Write $Y = tX$. Lemma 2.5 and the union bound show that the number of sources $n \leq Y$ for which some prime $p \leq P$ fails to divide $\sigma(n)$ is at most

$$O(\sqrt{Y}) + O\left(Y\pi(P) \exp\left[-c_1 \frac{\log \log Y}{P}\right]\right). \quad (5.6) \quad \boxed{\text{eq:sharp-bad-sour}}$$

The first term treats $p = 2$. The chosen range of P is permitted by Lemma 2.5 after decreasing ρ if necessary. Since

$$Ft = e^{(2-\varepsilon)W} = (\log \log X)^{o(1)}, \quad \exp\left[-c_1 \frac{\log \log Y}{P}\right] = (\log \log X)^{-c_1/\rho+o(1)},$$

we may also ensure $c_1/\rho > 3$. Multiplying (5.6) by the at most F possible cofactors per source gives $o(X\Delta(P))$ target-cofactor pairs.

Consider a remaining P -rough target N admitting a witness $N = F_e(n/e)$ with $n \leq tN$ and $e \leq F$. Again $e = 1$ is impossible. For $e \geq 2$, put

$$L = \text{lcm}(\{r : r \mid n, r < e\}), \quad C = \sum_{\substack{r \mid L \\ r < e}} \frac{L}{r}, \quad R = \sigma(n) - N.$$

The divisors of L below e are exactly the divisors of n below e . By Lemma 2.2,

$$R = \frac{n}{L}C.$$

Since every prime $p \leq P$ divides $\sigma(n)$ but not N , the integer $R = (n/L)C$ is P -rough. Writing $u = n/L$, both u and C are therefore P -rough. Since $e \leq F < P$ and $e \mid n = Lu$, every prime-power divisor of e already divides L . Consequently

$$n = Lu, \quad \gcd(u, P\#) = 1, \quad \gcd(C, P\#) = 1, \quad e \mid L \mid \Lambda(F).$$

As before $C \geq L \geq 2$, so $C > P$. On the other hand,

$$C \leq \sigma(L) \leq L \prod_{p \leq F} \left(1 - \frac{1}{p}\right)^{-1} \ll L \log F.$$

There is therefore an absolute constant $C_0 > 0$ such that

$$L > \frac{P}{C_0 W}. \quad (5.7) \quad \boxed{\text{eq:moving-kernel-}}$$

Uniformly for $L \mid \Lambda(F)$,

$$\log L \leq \log \Lambda(F) \ll F = o(\log X), \quad \log(P\#) \ll P = o(\log X).$$

It follows that $tX/L \geq P\#$ for all sufficiently large X . Complete periods modulo $P\#$ consequently give

$$\#\{u \leq tX/L : \gcd(u, P\#) = 1\} \leq \frac{2tX\Delta(P)}{L}.$$

For each fixed L , the admissible cofactors satisfy $e \mid L$, so there are at most $\tau(L)$ possibilities. Hence (5.7) and Lemma 5.1 bound the entire remaining low-cofactor witness count by

$$\begin{aligned} 2tX\Delta(P) \sum_{\substack{L \mid \Lambda(F) \\ L > P/(C_0 W)}} \frac{\tau(L)}{L} &\leq X\Delta(P) \exp\{-\varepsilon W + o(W)\} \\ &= o(X\Delta(P)). \end{aligned}$$

Choose a fixed integer $k \geq 2$ so large that $(k+1)\varepsilon > 2$. Lemma 2.6 bounds the targets with a witness $n \leq tN$ and cofactor $e > F$ by

$$C_k t^{k+1} F^{1-k} X.$$

After division by $X\Delta(P)$, this is at most

$$O_k(J \exp\{[2 - (k+1)\varepsilon + o(1)]W\}) = o(1).$$

Thus a positive proportion of the squarefree targets in (5.5) has no representation with $n \leq tN$. These targets are odd. Lemma 3.1 discards only $o(X/\log \log \log X) = o(X\Delta(P))$ further targets as unrepresented. Consequently a positive proportion still survives, without any use of the finite representability computation. Finally,

$$\log t = \left(\frac{1-\varepsilon}{\sqrt{2}} + o(1) \right) \sqrt{(\log \log \log X)(\log \log \log \log X)}.$$

Choose ε sufficiently small in terms of η . For $X/2 < N \leq X$, replacing X by N in these iterated logarithms changes the right-hand side by $o(1)$ relative to its main term. This proves (1.3).

For (1.4), apply the first assertion with a sufficiently small auxiliary value of η , raise the resulting ratios to the power s , and note that

$$\log \log \log \log X = o\left(\sqrt{(\log \log \log X)(\log \log \log \log X)}\right).$$

The factor $c_\eta/\log \log \log X$ in the proportion of surviving targets can therefore be absorbed into the prescribed exponent loss η . $\square$

6. BOUNDED REPRESENTATIONS AND COFACTOR RANGES

(sec:tightness)

We now study the representations themselves and their contribution to the distribution of the least representing integer. For $t > 0$ and $N \geq 1$, let

$$\begin{aligned} r_t(N) &= \#\{(e, d) \in \mathbb{N}^2 : F_e(d) = N, ed \leq tN\}, \\ r_t^*(N) &= \#\{(e, d) \in \mathbb{N}^2 : e \geq 2, F_e(d) = N, ed \leq tN\}. \end{aligned}$$

The second multiplicity counts proper divisor-prefix sums, excluding the full divisor sum. Equivalently, $r_t(N)$ counts the pairs (n, j) for which $N = \sigma_j(n)$ and $n \leq tN$. Thus, on the represented set $\mathcal{R}$, one has $r_t(N) > 0$ if and only if $f(N) \leq tN$.

Recall the empirical measures ν_X and the normalization $R(X)$ from (1.5). A mass at ∞ in a compactified subsequential limit records escape to infinity. Tightness on the finite half-line means

$$\lim_{T \rightarrow \infty} \sup_{X \geq 1} \nu_X((T, \infty]) = 0.$$

The associated density-one question is

$$\lim_{A \rightarrow \infty} \limsup_{X \rightarrow \infty} \frac{1}{X} \#\{N \leq X : r_A(N) = 0\} = 0. \quad (\text{T}) \quad \boxed{\text{eq:T}}$$

The order of the limits matters: A is fixed before $X \rightarrow \infty$. The next proposition shows that every fixed residue class has many representation pairs. This does not by itself control the number of distinct targets.

6.1. **A first moment in every residue class.** For fixed $Q \geq 1$ and $\rho \pmod{Q}$, put

$$\lambda_{Q,\rho}(A) = \liminf_{X \rightarrow \infty} \frac{Q}{X} \sum_{\substack{N \leq X \\ N \equiv \rho \pmod{Q}}} r_A(N).$$

lass-first-moment **Proposition 6.1.** *For every fixed $Q \geq 1$ and every residue class $\rho \pmod{Q}$,*

$$\lambda_{Q,\rho}(A) \gg_{Q,\rho} \frac{A}{(\log A)^2} \quad (A \rightarrow \infty).$$

In particular, this lower class mean tends to infinity in every residue class.

Proof. Fix Q and ρ , and set $d_0 = \gcd(\rho, Q)$. We first choose a fixed integer h satisfying

$$\gcd(\sigma(h), Q) = d_0. \tag{6.2} \quad \boxed{\text{eq:h-gcd}}$$

For each prime power $\ell^k \parallel d_0$, Dirichlet's theorem on primes in arithmetic progressions permits the choice of a different auxiliary prime q_ℓ congruent to 1 modulo every prime dividing Q . When $\ell = 2$, we also require $q_2 \equiv 1 \pmod{4}$. Put

$$h_\ell = q_\ell^{\ell^k - 1}, \quad h = \prod_{\ell^k \parallel d_0} h_\ell,$$

with $h = 1$ when $d_0 = 1$. The lifting-the-exponent formula gives $v_\ell(\sigma(h_\ell)) = k$. If r is a prime divisor of Q with $r \neq \ell$, then $\sigma(h_\ell) \equiv \ell^k \not\equiv 0 \pmod{r}$. Multiplicativity of σ proves (6.2).

The elementary criterion for a linear congruence now gives a unit $u_\rho \pmod{Q}$ for which

$$\sigma(h)u_\rho \equiv \rho \pmod{Q}. \tag{6.3} \quad \boxed{\text{eq:u-class}}$$

This includes the class $\rho = 0$. Prime-power by prime-power, divide by the common valuation in (6.2), solve for a unit modulo the remaining power, and combine the solutions by the Chinese remainder theorem.

Dirichlet's theorem also supplies infinitely many primes e satisfying

$$e > \max\{2, h, Q\}, \quad e \equiv -1 \pmod{Q}.$$

For each such prime e , define

$$\mathcal{S}_e(u_\rho) = \{t \geq 1 : t \equiv u_\rho \pmod{Q}, \gcd(t, e\#) = 1\}.$$

The Chinese remainder theorem shows that this periodic set has density

$$\delta_e = \frac{1}{Q} \prod_{\substack{p \leq e \\ p \nmid Q}} \left(1 - \frac{1}{p}\right).$$

Every prime factor of $t \in \mathcal{S}_e(u_\rho)$ is larger than e .

For $t \in \mathcal{S}_e(u_\rho)$, the divisors of eht below e are exactly the divisors of h . Complementary divisors therefore give

$$F_e(ht) = \sigma(h)H_e(t), \quad H_e(t) = (e+1)\sigma(t) - et = t + (e+1)s(t), \tag{6.4} \quad \boxed{\text{eq:He}}$$

where $s(t) = \sigma(t) - t$. Since $Q \mid e+1$, equations (6.3) and (6.4) give

$$F_e(ht) \equiv \sigma(h)t \equiv \rho \pmod{Q}.$$

The corresponding representing integer is $n = eht$. Since $H_e(t) \geq t$,

$$\frac{n}{F_e(ht)} \leq \frac{eh}{\sigma(h)} \leq eh. \tag{6.5} \quad \boxed{\text{eq:representing-r}}$$

We next bound the represented value. Complementary divisors give

$$\frac{H_e(t)}{t} = 1 + (e+1) \sum_{\substack{a \mid t \\ a > 1}} \frac{1}{a}.$$

For a fixed integer $a > 1$ with $\gcd(a, e\#) = 1$, the Chinese remainder theorem shows that the set of $t \in \mathcal{S}_e(u_\rho)$ divisible by a has density δ_e/a . The contribution of a divisor a to the normalized average is at most a^{-2} , since at most T/a integers up to T are divisible by a . Thus $\sum_{a \geq 1} a^{-2} < \infty$ justifies interchanging the limit and the divisor sum. Therefore

$$\begin{aligned} \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{\substack{t \leq T \\ t \in \mathcal{S}_e(u_\rho)}} \frac{H_e(t)}{t} &= \delta_e \left(1 + (e+1) \sum_{\substack{a > 1 \\ \gcd(a, e\#) = 1}} \frac{1}{a^2} \right) \\ &\leq 3\delta_e. \end{aligned}$$

Every integer $a > 1$ coprime to $e\#$ exceeds e . Hence

$$(e+1) \sum_{\substack{a > 1 \\ \gcd(a, e\#) = 1}} \frac{1}{a^2} \leq (e+1) \sum_{a > e} \frac{1}{a^2} \leq \frac{e+1}{e} \leq \frac{3}{2},$$

which proves the last inequality. Markov's inequality shows that

$$\mathcal{G}_e = \{t \in \mathcal{S}_e(u_\rho) : H_e(t) \leq 6t\}$$

has lower density at least $\delta_e/2$.

For a fixed A , restrict to primes e satisfying

$$\max\{2, h, Q\} < e \leq A/h, \quad e \equiv -1 \pmod{Q}.$$

If $t \in \mathcal{G}_e$ and $t \leq X/(6\sigma(h))$, then the represented value in (6.4) is at most X , lies in the class ρ , and has a representing integer $n \leq AN$ by (6.5). Distinct pairs (e, t) give distinct representing integers. Indeed, if $e < e'$ and $eht = e'ht'$, then $e \mid t'$, contrary to the fact that every prime factor of t' exceeds e' . If $e = e'$, equality of the representing integers forces $t = t'$. Hence these pairs give distinct representation pairs (n, j) , even if some represented values coincide. For each eligible prime e , the number of such $t \leq X/(6\sigma(h))$ is at least $(\delta_e/2 + o(1))X/(6\sigma(h))$. Multiplying by Q/X and summing over e gives

$$\lambda_{Q, \rho}(A) \geq \frac{1}{12\sigma(h)} \sum_{\substack{\max\{2, h, Q\} < e \leq A/h \\ e \text{ prime} \\ e \equiv -1 \pmod{Q}}} \prod_{\substack{p \leq e \\ p \nmid Q}} \left(1 - \frac{1}{p}\right). \quad (6.6) \quad \boxed{\text{eq:class-lower}}$$

Mertens' product theorem gives

$$\prod_{\substack{p \leq e \\ p \nmid Q}} \left(1 - \frac{1}{p}\right) \sim \frac{Q}{\varphi(Q)} \frac{e^{-\gamma}}{\log e}.$$

For $Z \geq 2$, the prime number theorem in arithmetic progressions, followed by partial summation, gives, as $Z \rightarrow \infty$,

$$\sum_{\substack{e \leq Z \\ e \text{ prime} \\ e \equiv -1 \pmod{Q}}} \frac{1}{\log e} \sim \frac{Z}{\varphi(Q)(\log Z)^2}.$$

The omitted primes below the fixed threshold $\max\{2, h, Q\}$ do not change this asymptotic. Taking $Z = A/h$ in (6.6) proves the proposition. $\square$

Remark 6.2. Proposition 6.1 counts representation pairs. A small set of represented values could carry many pairs, so the proposition alone gives no bound for $\#\{N \leq X : r_A(N) = 0\}$. Thus the proposition does not by itself imply (T).

6.2. Prime cofactor ranges. The exact forced modulus of Section 2 always contains the cofactor when that cofactor is prime. This gives a simple upper bound for the density of its range.

Corollary 6.3. *Let p be prime. Then $F_p(d) \equiv \sigma(pd) \pmod{p}$ for every $d \geq 1$, and*

$$\#(F_p(\mathbb{N}) \cap [1, X]) \leq \frac{X}{p} + o_p(X), \quad \text{so} \quad \bar{d}(F_p(\mathbb{N})) \leq \frac{1}{p}.$$

Proof. Every $j \in D$ satisfies $j < p$, so $p \nmid M$ and $g = 1$. Hence $p \mid K_{p,d}$, and the congruence follows from (2.3). Now let $N = F_p(d) \leq X$ with $p \nmid N$. Then $p \nmid \sigma(pd)$, and $pd \leq pN \leq pX$ because $d \leq F_p(d)$. Distinct N have distinct sources pd , so these N number at most $\#\{n \leq pX : p \nmid \sigma(n)\}$, which is $o_p(X)$ by Lemma 2.3. Every other value of F_p up to X is a multiple of p . $\square$

For $p = 2$ the bound $1/2$ is weaker than Proposition 6.6 below. For every prime p , the numerical upper bound is $1/p$; the error term $o_p(X)$ is asserted only with p fixed.

6.3. Bounded cofactors. For a positive integer A , put

$$G_A = \bigcup_{e \leq A} F_e(\mathbb{N}), \quad \Lambda(A) = \text{lcm}(1, \dots, A), \quad P_A = A\Lambda(A).$$

Every $N = F_e(d)$ with $e \leq A$ has a representing integer $n = ed \leq AN$, because $F_e(d) \geq d$. Thus

$$G_A \subseteq \{N : r_A(N) > 0\}.$$

The upper-tail estimate shows that every fixed union of cofactor ranges misses a positive-density set, even among squarefree targets.

Corollary 6.4. *For every fixed $A \geq 1$,*

$$\underline{d}(\mathbb{N} \setminus G_A) > 0.$$

More precisely, as $A \rightarrow \infty$,

$$\underline{d}\{N : N \text{ squarefree}, N \notin G_A\} \geq \left(\frac{e^{-\gamma}}{2} - o(1)\right) \frac{\log \log \log A}{(\log A)(\log \log A)}. \quad (6.7) \quad \boxed{\text{eq:fixed-cofactor}}$$

Proof. If $N \in \mathcal{R}$ and $f(N) > AN$, then $N \notin G_A$. The quantitative assertion follows from Theorem 1.3 with $T = A$. For any fixed smaller A , apply that theorem with one sufficiently large fixed $T \geq A$. $\square$

Write

$$\delta_A = \Delta(P_A), \quad \eta_A = \bar{d}\{N : \gcd(N, P_A\#) > 1, N \notin G_A\}.$$

For $2 \leq e \leq A$, every forced modulus $K_{e,d}$ has a prime factor at most P_A . Indeed, if $e/g > 1$, use a prime factor of e/g ; otherwise $K_{e,d} = C$, where $2 \leq C \leq (e-1)M \leq P_A$. There are only finitely many such moduli. Lemma 2.2 and fixed-modulus normality therefore imply that all but $o_A(X)$ values $F_e(d) \leq X$ have a prime factor at most P_A : their sources satisfy $ed \leq AX$, and a fixed source contributes at most A values. The case $e = 1$ is covered by Lemma 2.4. Hence

$$\#\{N \leq X : \gcd(N, P_A\#) = 1, N \in G_A\} = o_A(X).$$

Thus G_A has density zero among the integers coprime to $P_A\#$. Hence the coprime part of $\mathbb{N} \setminus G_A$ has density δ_A . Since the coprime and noncoprime parts are disjoint,

$$1 - \underline{d}(G_A) = \delta_A + \eta_A. \quad (6.8) \quad \boxed{\text{eq:bounded-cofact}}$$

Moreover, Mertens' theorem and the prime number theorem, in the form $\psi(A) = A + o(A)$, give $\log P_A = \log A + \psi(A) = A + o(A)$ and hence

$$\delta_A = \frac{e^{-\gamma} + o(1)}{A}. \quad (6.9) \quad \boxed{\text{eq:delta-A-asympt}}$$

The two terms in (6.8) are of different orders, and it is η_A that dominates. Recall the notation

$$L(A) = \frac{\log \log \log A}{(\log A)(\log \log A)}.$$

Proposition 6.5. *As $A \rightarrow \infty$,*

$$\eta_A \geq \left(\frac{e^{-\gamma}}{2} - o(1) \right) L(A), \quad \text{so} \quad A\eta_A \rightarrow \infty.$$

Proof. By (6.7), $1 - \underline{d}(G_A) \geq (e^{-\gamma}/2 - o(1))L(A)$. Since $AL(A) \rightarrow \infty$, (6.9) gives $\delta_A = o(L(A))$. Subtracting in (6.8) proves the first claim, and the second follows from $AL(A) \rightarrow \infty$. $\square$

One might guess that coprimality to $P_A\#$ is asymptotically the only obstruction, so that $1 - \underline{d}(G_A) \sim e^{-\gamma}/A$. The proposition rules this out. The integers missed by G_A for other reasons outnumber the coprime ones by a factor that tends to infinity.

Now set

$$\theta_2 = \bar{d}(F_2(\mathbb{N})).$$

$\langle \text{prop:theta-two} \rangle$ **Proposition 6.6.** *One has*

$$\theta_2 \leq \frac{1}{2} - 0.0602757 = 0.4397243.$$

Proof. The only divisor of $2d$ exceeding d is $2d$, and hence

$$F_2(d) = \sigma(2d) - 2d = s(2d). \quad (6.10) \quad \boxed{\text{eq:F2-aliquot}}$$

Thus $F_2(\mathbb{N}) \subseteq s(\mathbb{N})$ and the cofactor-two range omits every untouchable number. Write

$$\mathcal{U} = \mathbb{N} \setminus s(\mathbb{N}).$$

We first note that the odd part of $F_2(\mathbb{N})$ is negligible. If n is even, then $s(n)$ is odd precisely when $\sigma(n)$ is odd, which occurs precisely when n is a square or twice a square. For even $n \geq 4$ one also has $s(n) \geq 1 + n/2$; the exceptional case $n = 2$ is harmless. Thus $s(n) \leq X$ implies $n < 2X$, apart from at most this one value. Consequently,

$$\#\{N \leq X : N \text{ odd}, N \in F_2(\mathbb{N})\} = O(\sqrt{X}). \quad (6.11) \quad \boxed{\text{eq:F2-odd-negligi}}$$

Almost all untouchable numbers are even. Indeed, let N be odd. If $N - 1 = p + q$ with distinct primes p and q , then the proper divisors of pq are $1, p, q$, and therefore $s(pq) = N$. The exceptional-set theorem of Montgomery and Vaughan shows that the even integers up to X that are not a sum of two primes number $O(X^{1-\kappa})$ for some $\kappa > 0$; the diagonal possibilities $N - 1 = 2p$ contribute $O(X/\log X)$ more [13]. Hence

$$\#\{N \leq X : N \text{ odd}, N \in \mathcal{U}\} = o(X). \quad (6.12) \quad \boxed{\text{eq:odd-untouchabl}}$$

Chen and Zhao proved [3] that

$$\underline{d}(\mathcal{U}) \geq 0.0602757.$$

By (6.12), the number of even untouchables not exceeding X is at least $(0.0602757 + o(1))X$. There are $X/2 + O(1)$ even integers up to X , and by (6.11) all but $O(\sqrt{X})$ elements of $F_2(\mathbb{N}) \cap [1, X]$ are even. Since (6.10) shows that this range misses every member of $\mathcal{U}$, we obtain

$$\#(F_2(\mathbb{N}) \cap [1, X]) \leq \frac{X}{2} - 0.0602757X + o(X).$$

Taking the upper limit proves the proposition. $\square$

Corollary 6.7. *For the residual density defined above,*

$$\eta_2 \geq \frac{2}{3} - 0.4397243 > 0.2269.$$

Proof. Since $F_1(d) = \sigma(d)$,

$$G_2 = \sigma(\mathbb{N}) \cup F_2(\mathbb{N}).$$

Lemma 2.4 implies that $\sigma(\mathbb{N})$ has density zero, and therefore $\bar{d}(G_2) \leq \theta_2$. Moreover, $\Lambda(2) = 2$, $P_2 = 4$, $P_2\# = 6$, and so $\delta_2 = 1/3$. Using (6.8),

$$\eta_2 = 1 - \underline{d}(G_2) - \delta_2 \geq \frac{2}{3} - \bar{d}(G_2) \geq \frac{2}{3} - 0.4397243 > 0.2269.$$

□

We conjecture that bounded cofactors cover sets of lower density tending to one as the cofactor bound grows.

Conjecture 6.8. *As $A \rightarrow \infty$ through the positive integers,*

$$\underline{d}(G_A) \rightarrow 1.$$

Equivalently, $\eta_A = o(1)$.

The equivalence in Conjecture 6.8 follows from (6.8) and (6.9).

Corollary 5.3 shows where the congruence argument stops. The set V_A consists of A -rough integers, so the forced congruences say nothing about integers with a prime factor at most A that are missed by G_A . The A -rough part of $\mathbb{N} \setminus G_A$ has upper density at most $\Delta(A)$, which tends to zero. Conjecture 6.8 is therefore equivalent to the assertion that the part of $\mathbb{N} \setminus G_A$ with a prime factor at most A has upper density tending to zero. Every element of that part satisfies some forced congruence and is nevertheless not a value.

The conjectured density-one coverage by bounded cofactors is equivalent to tightness of the empirical ratio distributions.

Proposition 6.9. *The empirical measures ν_X form a tight family on $[0, \infty)$ if and only if*

$$\lim_{E \rightarrow \infty} \bar{d}(\mathbb{N} \setminus G_E) = 0. \quad (6.13) \quad \boxed{\text{eq:tightness-crit}}$$

Equivalently,

$$\lim_{E \rightarrow \infty} \underline{d}(G_E) = 1.$$

Thus Conjecture 6.8 is exactly the assertion that the family $\{\nu_X\}$ is tight.

Proof. The summand d occurs in $F_e(d)$, so $F_e(d) \geq d$. Hence

$$N \in G_E \implies f(N) \leq ed \leq EN \quad \text{for some } e \leq E.$$

Using (1.1) to replace the normalization by $X + O(1)$, we obtain

$$\limsup_{X \rightarrow \infty} \nu_X((E, \infty]) \leq \bar{d}(\mathbb{N} \setminus G_E).$$

Thus (6.13) makes the tail uniformly small for all sufficiently large X . For X in a bounded interval, only finitely many represented targets occur, so their ratios have a common finite upper bound. This proves tightness for the whole family.

Conversely, fix $T \geq 1$ and an integer $k \geq 2$. Every represented $N \notin G_E$ with $f(N) \leq TN$ has a witnessing cofactor $e > E$. Lemma 2.6 therefore gives

$$\bar{d}(\mathbb{N} \setminus G_E) \leq \limsup_{X \rightarrow \infty} \nu_X((T, \infty]) + C_k T^{k+1} E^{1-k}.$$

If the family is tight, first let $E \rightarrow \infty$ and then let $T \rightarrow \infty$. The right-hand side tends to zero. Finally, $\bar{d}(\mathbb{N} \setminus G_E) = 1 - \underline{d}(G_E)$ identically. □

Since $r_A(N) > 0$ is equivalent, apart from the two unrepresented integers, to $f(N) \leq AN$, Proposition 6.9 also shows that the density-one statement (T) is equivalent to the weak bounded-cofactor conjecture.

7. WITNESS MEASURES AND LIMITING LAWS

For a residue class $a \pmod{\Lambda(e)}$, put

$$c_e(a) = \sum_{\substack{1 \leq j < e \\ j|a}} \frac{1}{j}.$$

Write $\mathcal{D}$ for the statistical law of $h(n) = \sigma(n)/n$. The Davenport theorem [4] and its version for arithmetic progressions by Pollack [15, Lemma 2] gives an atomless measure $\nu_{a,e}$ characterized by

$$\nu_{a,e}(B) = \lim_{Y \rightarrow \infty} \frac{1}{Y} \#\{n \leq Y : n \equiv a \pmod{\Lambda(e)}, h(n) \in B\}$$

for every continuity set B . In particular, each $\nu_{a,e}$ has total mass $1/\Lambda(e)$, and

$$\sum_{a \bmod \Lambda(e)} \nu_{a,e} = \mathcal{D}, \quad 0 \leq \nu_{a,e} \leq \mathcal{D}. \quad (7.1) \quad \text{eq:dadd:progressi}$$

The progression laws yield explicit continuous limits for the mean numbers of bounded representations.

add:witness-means

Proposition 7.1. *For each $t > 0$ the limits*

$$W(t) = \lim_{X \rightarrow \infty} \frac{1}{X} \sum_{N \leq X} r_t(N), \quad W^*(t) = \lim_{X \rightarrow \infty} \frac{1}{X} \sum_{N \leq X} r_t^*(N)$$

exist, are finite, and are continuous. More explicitly,

$$\begin{aligned} W(t) &= \sum_{e \geq 1} w_e(t), \quad W^*(t) = \sum_{e \geq 2} w_e(t), \\ w_e(t) &= \sum_{\substack{a \bmod \Lambda(e) \\ e|a}} \int_{h - c_e(a) \geq 1/t} \frac{d\nu_{a,e}(h)}{h - c_e(a)}. \end{aligned}$$

Both series converge uniformly for $0 < t \leq T$, for every fixed $T > 0$, and $W(t) \rightarrow 0$ as $t \downarrow 0$.

Proof. For $n \equiv a \pmod{\Lambda(e)}$ with $e \mid a$, complementary divisors give the exact identities

$$F_e(n/e) = ng_e(n), \quad g_e(n) = h(n) - c_e(a).$$

In particular $g_e(n) \geq 1/e$. A witness with target at most X and ratio at most t is characterized by

$$g_e(n) \geq 1/t, \quad ng_e(n) \leq X.$$

For completeness, the progression theorem also yields the joint limit

$$\frac{1}{X} \sum_{\substack{n \leq tX \\ n \equiv a \pmod{\Lambda(e)}}} \delta_{(n/X, h(n))} \implies \mathbf{1}_{[0,t]}(u) du d\nu_{a,e}(h).$$

Indeed, on a rectangle $(\alpha, \beta] \times B$ the assertion follows by subtracting the progression counts at βX and αX . The limiting law in h is tight, so rectangle approximation proves the joint weak convergence. The boundary of $\{h - c_e(a) \geq 1/t, u(h - c_e(a)) \leq 1\}$ has zero product measure: the horizontal boundary has zero measure by atomlessness, and for each $h > c_e(a)$ the curved boundary contains only one value of u . The support condition $h - c_e(a) \geq 1/e$

holds for $\nu_{a,e}$ because it holds for every source in the progression. We may therefore pass to the limit in the witness condition. Tonelli's theorem gives

$$\begin{aligned} & \lim_{X \rightarrow \infty} \frac{1}{X} \# \{n \equiv a \pmod{\Lambda(e)} : g_e(n) \geq 1/t, \, n g_e(n) \leq X\} \\ &= \int_0^t \nu_{a,e} \{h : h - c_e(a) \geq 1/t, \, h - c_e(a) \leq 1/u\} du \\ &= \int_{h - c_e(a) \geq 1/t} \frac{d\nu_{a,e}(h)}{h - c_e(a)}. \end{aligned}$$

For the cofactor tail, Lemma 2.6 counts witnessing pairs themselves:

$$\begin{aligned} \frac{1}{X} \sum_{N \leq X} \# \{(e, d) : e > E, \, F_e(d) = N, \, ed \leq tN\} &\leq \frac{t^k}{X} \sum_{n \leq tX} \sum_{\substack{e|n \\ e > E}} g_e(n)^k \\ &\leq C_k t^{k+1} E^{1-k}. \end{aligned}$$

This proves both limits and uniform convergence on $0 < t \leq T$. Each fixed $w_e(t)$ is continuous by atomlessness, and tends to zero with t since its integrand is bounded by t on its domain. The uniform tail proves all remaining assertions. $\square$

Proposition 6.1 with $Q = 1$ also shows that $W(t) \gg t/(\log t)^2$ as $t \rightarrow \infty$. Since $w_1(t) \leq 1$, the same lower bound holds for $W^*(t)$.

7.1. Every subsequential finite law is singular continuous. All subsequential limits of the empirical ratio distributions have atomless, nonzero finite parts carried by a single Lebesgue-null set.

ersal-singularity **Theorem 7.2.** *There is a fixed σ -compact, Lebesgue-null set $\mathcal{Z} \subset (0, \infty)$ such that every weak subsequential limit ν of ν_X on $[0, \infty]$ satisfies*

$$\begin{aligned} 0 &\in \text{supp}(\nu), \quad \nu(\{0\}) = 0, \\ \nu((0, \infty) \setminus \mathcal{Z}) &= 0, \quad \nu(\{x\}) = 0 \quad (0 < x < \infty). \end{aligned}$$

Hence its restriction to $[0, \infty)$ is a nonzero singular continuous measure. If (6.13) holds, every subsequential limit is a singular continuous probability law on $[0, \infty)$; in particular, none admits a Lebesgue density. Neither uniqueness of the limit nor tightness is asserted unconditionally.

Proof. Erdős's singularity theorem asserts that the continuous Davenport law $\mathcal{D}$ is purely Lebesgue singular; see the account in [6, p. 59]. By regularity there is a σ -compact Lebesgue-null set $K \subset [1, \infty)$ with $\mathcal{D}(K) = 1$. For $c \in \mathbb{R}$, write

$$\psi_c(h) = \frac{1}{h - c} \quad (h > c),$$

and define the positive witness measure

$$\mathcal{W} = \sum_{e \geq 1} \sum_{\substack{a \bmod \Lambda(e) \\ e|a}} (\psi_{c_e(a)})_* (\mathbf{1}_{\{h > c_e(a)\}} \psi_{c_e(a)}(h) d\nu_{a,e}(h)).$$

Proposition 7.1 gives

$$\mathcal{W}((0, t]) = W(t) < \infty \quad (t > 0),$$

so $\mathcal{W}$ is locally finite and atomless. By (7.1), every $\nu_{a,e}$ is carried by K . Therefore $\mathcal{W}$ is carried by

$$\mathcal{Z} = \bigcup_{e \geq 1} \bigcup_{\substack{a \bmod \Lambda(e) \\ e|a}} \bigcup_{j \geq 1} \psi_{c_e(a)}(K \cap [c_e(a) + 1/j, j]).$$

Each map in the last display is Lipschitz on its indicated set. Thus $\mathcal{Z}$ is σ -compact and Lebesgue null.

At finite scale, define the empirical witness measure

$$\mathcal{W}_X = \frac{1}{X} \sum_{N \leq X} \sum_{\substack{e, d \geq 1 \\ F_e(d) = N}} \delta_{ed/N}.$$

On every bounded ratio interval it is finite. Proposition 7.1 and continuity of W show that $\mathcal{W}_X \rightarrow \mathcal{W}$ vaguely on $(0, \infty)$. For every nonnegative $\varphi \in C_c((0, \infty))$, choosing a minimizing witness separately for each represented target gives

$$\int \varphi d\nu_X \leq \frac{X}{R(X)} \int \varphi d\mathcal{W}_X.$$

Along any weakly convergent subsequence $\nu_{X_j} \Rightarrow \nu$, pass to the limit to obtain

$$\int \varphi d\nu \leq \int \varphi d\mathcal{W}.$$

Hence $\nu|_{(0, \infty)} \leq \mathcal{W}$. This proves both singularity on the common carrier and the absence of positive finite atoms.

Finally, the same witness domination and portmanteau give, for every $t > 0$,

$$\nu(\{0\}) \leq \nu([0, t]) \leq \liminf_{j \rightarrow \infty} \nu_{X_j}([0, t]) \leq W(t).$$

Let $t \downarrow 0$ and use Proposition 7.1. To see that zero nevertheless belongs to the support, fix $\delta > 0$. Theorem 1.2 and the closed-set direction of portmanteau give

$$\nu([0, \delta]) \geq \limsup_{j \rightarrow \infty} \nu_{X_j}([0, \delta]) > 0.$$

Applying this with arbitrarily small δ proves the support claim and shows that the finite part is nonzero. Finally, tightness excludes mass at ∞ in every compactified subsequential limit, so Proposition 6.9 gives the last assertion. $\square$

The tail estimates imply divergence of positive and logarithmic moments, while inverse moments remain finite and pass to subsequential limits.

Corollary 7.3. *Every compactified subsequential law ν satisfies, as $T \rightarrow \infty$,*

$$\nu((T, \infty]) \geq \left(\frac{e^{-\gamma}}{2} - o(1) \right) \frac{\log \log \log T}{(\log T)(\log \log T)}. \quad (7.2) \quad \boxed{\text{eq:dadd:subsequen}}$$

Moreover, for every $s > 0$,

$$\lim_{X \rightarrow \infty} \int x^s d\nu_X(x) = \infty, \quad \int_{[0, \infty]} x^s d\nu(x) = \infty.$$

The logarithmic means also satisfy

$$\lim_{X \rightarrow \infty} \int \log x d\nu_X(x) = \infty, \quad \int_{[0, \infty]} \log x d\nu(x) = \infty.$$

In contrast, for every $s > 0$,

$$\sup_{X \geq 1} \int x^{-s} d\nu_X(x) < \infty, \quad \int x^{-s} d\nu(x) < \infty,$$

where $\infty^{-s} = 0$. Along every weakly convergent subsequence, the inverse moments converge to those of the limiting law. In particular, the geometric means of $f(N)/N$ tend to infinity; even assuming tightness, every subsequential probability law has infinite logarithmic mean and infinite moments of all positive orders.

Proof. The closed-set direction of portmanteau and Theorem 1.3 give

$$\nu([T, \infty]) \geq \limsup_{j \rightarrow \infty} \nu_{X_j}([T, \infty]) \geq \liminf_{X \rightarrow \infty} \nu_X([T, \infty]).$$

Theorem 7.2 excludes an atom at T , proving (7.2). For either ν_X or ν , the s th moment is at least T^s times its mass on $(T, \infty]$. First take the lower limit in X when appropriate, and then let $T \rightarrow \infty$ in the preceding lower bound.

For the logarithmic means, layer cake and Fatou's lemma give

$$\begin{aligned} \liminf_{X \rightarrow \infty} \int \log^+ x \, d\nu_X(x) &\geq \int_{T_0}^B \liminf_{X \rightarrow \infty} \nu_X([T, \infty]) \frac{dT}{T} \\ &\geq \left(\frac{e^{-\gamma}}{4} - o(1) \right) (\log \log \log B)^2 \quad (B \rightarrow \infty). \end{aligned}$$

The same calculation using (7.2) shows $\int \log^+ x \, d\nu(x) = \infty$. On the other hand, Theorem 1.1 and the bound $X/R(X) \leq 3$ give, for some $c > 0$ and all sufficiently large u ,

$$\sup_X \nu_X([0, e^{-u}]) \ll \exp(-\exp(e^{cu})).$$

Layer cake therefore bounds $\int \log^- x \, d\nu_X(x)$ uniformly in X . To pass the bound to ν , apply weak convergence to the continuous bounded functions $\min\{M, \log^- x\}$ on $[0, \infty]$, defined to equal M at zero and zero at infinity, and then let $M \rightarrow \infty$. Subtracting the bounded negative parts proves both logarithmic assertions.

Finally, the same uniform lower-tail bound and layer cake show that

$$\sup_{X \geq 1} \int_{\{x < 1\}} x^{-s} \, d\nu_X(x) \leq 1 + s \int_0^\infty e^{su} \sup_{X \geq 1} \nu_X([0, e^{-u}]) \, du < \infty.$$

Apply this also with $2s$ in place of s to obtain uniform integrability of x^{-s} . The bounded truncations $\min\{M, x^{-s}\}$ are continuous on the compactified half-line when assigned their limiting endpoint values. Weak convergence, followed by removal of the truncation, proves the claimed convergence and finiteness of inverse moments. $\square$

7.2. The collision obstruction. Convergence of the empirical ratio distributions is equivalent to convergence of the surplus caused by multiple proper witnesses for the same target.

(collision-criterion) **Proposition 7.4.** *Put*

$$K_X(t) = \frac{1}{X} \sum_{N \leq X} (r_t^*(N) - 1)_+.$$

For every fixed $t > 0$,

$$\nu_X([0, t]) = W^*(t) - K_X(t) + o(1). \tag{7.3} \text{eq:dadd:collision}$$

Consequently, the threshold density exists if and only if the genuine proper-witness collision surplus $K_X(t)$ converges. The whole sequence ν_X converges weakly on $[0, \infty]$ if and only if $K_X(t)$ converges for every positive rational t .

Proof. A represented target absent from the support of r_t^* can only be represented at threshold t using cofactor $e = 1$; such a target belongs to $\text{range}(\sigma)$. Lemma 2.4 therefore shows that

$$\nu_X([0, t]) = \frac{1}{X} \#\{N \leq X : r_t^*(N) > 0\} + o(1).$$

For every nonnegative integer r , $\mathbf{1}_{\{r > 0\}} = r - (r - 1)_+$. Sum this identity over $N \leq X$ with $r = r_t^*(N)$ and apply Proposition 7.1.

For the final assertion, all compactified subsequential laws have no finite atoms by Theorem 7.2. If the collision surplus converges at each positive rational threshold, the displayed identity forces any two subsequential laws to agree on $[0, t]$ at all those thresholds, hence everywhere, including at infinity. Compactness then gives convergence of the full sequence. Conversely,

weak convergence implies convergence of the mass of each $[0, t]$, because its boundary has zero limiting measure, and the identity gives convergence of $K_X(t)$. $\square$

This collision surplus is bounded away from zero once the ratio threshold is at least two.

Proposition 7.5. *For every $t \geq 2$,*

$$\liminf_{X \rightarrow \infty} K_X(t) > \frac{1}{9}.$$

Proof. Put $Q = 5\# = 30$, so that $\Delta(5) = \varphi(Q)/Q = 4/15$, and let

$$\mathcal{C} = \{2, 4, 8, 10\}.$$

For $D \in \mathcal{C}$ and $\gcd(u, Q) = 1$, consider the canonical cofactor-two witness

$$N = s(Du) = \sigma(D)\sigma(u) - Du = u(s(D) + \sigma(D)(h(u) - 1)). \quad (7.4) \quad \boxed{\text{eq:dadd:collision}}$$

Its source $n = Du$ is even, so $N = s(n) \geq n/2$ and its source-to-target ratio is at most $2 \leq t$.

Let ν_Q be the unnormalized Davenport law restricted to $\gcd(u, Q) = 1$, of total mass $\Delta(5)$. Expanding $h(u)$ as a divisor sum gives

$$\frac{1}{\Delta(5)} \int (h - 1) d\nu_Q(h) = \sum_{\substack{d \geq 1 \\ \gcd(d, Q) = 1}} \frac{1}{d^2} - 1 = \frac{8\pi^2}{75} - 1 =: H.$$

The passage to this first moment is justified by the uniformly bounded second moment of $h(u)$, which follows from Lemma 4.1 with $q = 3$. The radial partition argument from Proposition 7.1 gives the source intensity

$$\begin{aligned} W_D &:= \lim_{X \rightarrow \infty} \frac{1}{X} \# \{u : \gcd(u, Q) = 1, s(Du) \leq X\} \\ &= \int \frac{d\nu_Q(h)}{s(D) + \sigma(D)(h - 1)} \geq \frac{\Delta(5)}{s(D) + \sigma(D)H}, \end{aligned}$$

where the last inequality is Jensen's inequality.

Choose the fixed modulus $q = 2^4 \cdot 3 \cdot 5^2$. Lemma 2.3 implies that, apart from $o(X)$ sources $n \leq 2X$, one has

$$v_p(\sigma(n)) > v_p(D) = v_p(n) \quad (p = 2, 3, 5).$$

For every remaining target in (7.4),

$$v_p(N) = v_p(\sigma(n) - n) = v_p(D) \quad (p = 2, 3, 5).$$

Thus, up to $o(X)$ exceptions, its image is contained in the cylinder

$$\mathcal{T}_D = \{Dv : \gcd(v, 30) = 1\}, \quad d(\mathcal{T}_D) = \frac{\Delta(5)}{D}.$$

If $r_D(N)$ denotes the number of parameters u giving N in (7.4), it follows that

$$\liminf_{X \rightarrow \infty} \frac{1}{X} \sum_{N \leq X} (r_D(N) - 1)_+ \geq \Delta(5) \left(\frac{1}{s(D) + \sigma(D)H} - \frac{1}{D} \right).$$

The four cores have distinct valuation vectors at 2, 3, 5, whereas u is coprime to 30. Thus their source families, and hence their cofactor-two witnessing pairs, are disjoint. The elementary inequality

$$\sum_{D \in \mathcal{C}} (r_D(N) - 1)_+ \leq \left(\sum_{D \in \mathcal{C}} r_D(N) - 1 \right)_+ \leq (r_t^*(N) - 1)_+$$

therefore allows the four lower bounds to be added.

Finally, $\pi < 22/7$ gives $H < 197/3675$. Since

$$(D, s(D), \sigma(D)) \in \{(2, 1, 3), (4, 3, 7), (8, 7, 15), (10, 8, 18)\},$$

all four terms are positive and exact rational arithmetic gives

$$\begin{aligned} \liminf_{X \rightarrow \infty} K_X(t) &> \frac{4}{15} \sum_{(D,s,B)} \left(\frac{1}{s + 197B/3675} - \frac{1}{D} \right) \\ &= \frac{3492203725711}{31001587618275} > \frac{1}{9}. \end{aligned}$$

This lower bound proves that the collision term in (7.3) is substantial; it does not prove that the term converges. $\square$

8. FURTHER QUESTIONS

The bounds at the two tails and the structural results above leave several questions about the typical size of the least representing integer.

Theorem 1.3 gives a fixed-level lower bound of order

$$\frac{\log \log \log T}{(\log T)(\log \log T)},$$

while Theorem 1.4, for each fixed $\eta > 0$, reaches

$$\exp \left[\left(\frac{1}{\sqrt{2}} - \eta \right) \sqrt{(\log \log \log N)(\log \log \log \log N)} \right]$$

on $\gg_\eta X / \log \log \log X$ targets. What are the correct fixed-level tail and the largest growing scale with a comparable counting estimate? In particular, does the constant $1/\sqrt{2}$ reflect the problem itself or the present estimate for the divisor-kernel tail? At the lower tail, it also remains to quantify the positive-density result as the ratio tends to zero.

Proposition 6.9 identifies tightness with $\underline{d}(G_E) \rightarrow 1$. The periodic obstruction V_E lies among the E -rough integers, whose density tends to zero. Tightness is therefore equivalent to the upper density tending to zero of those integers outside G_E that have a prime factor at most E . Such integers satisfy at least one of the forced divisibility conditions, but are still not values. The congruence arguments here do not control this residual set.

The witness means $W(t)$ and $W^*(t)$ exist, and Proposition 6.1 shows that the first moment of bounded representations grows in every fixed residue class. Nevertheless, coverage depends on how these representations are distributed among their targets. Proposition 7.4 gives an exact criterion at each threshold: convergence of the empirical distribution function is equivalent to convergence of the proper-witness collision surplus. The lower bound $\liminf K_X(t) > 1/9$ for $t \geq 2$ confirms that these collisions cannot be neglected. It provides no convergence result. A second-moment bound comparable to the square of the first moment could give positive-density coverage; density-one coverage would require an additional argument, for example a sufficiently small normalized variance for a suitable family of witnesses.

Every compactified subsequential law has a singular continuous finite part on the same null set, with zero in its support and no atom there. If tightness holds, the resulting probability laws have unbounded support, infinite logarithmic mean, and infinite moments of every positive order. It remains to prove or disprove tightness, convergence of the full sequence, uniqueness of its subsequential limits, and full support on $(0, \infty)$. The divergence of arithmetic and geometric means is compatible with tightness and answers none of these existence or uniqueness questions.

We have $\underline{d}(F_1(\mathbb{N})) = 0$, $\bar{d}(F_2(\mathbb{N})) \leq 0.4397243$, and $\bar{d}(F_p(\mathbb{N})) \leq 1/p$ for every prime p . Luca and Pomerance's work gives positive lower density for $F_2(\mathbb{N})$ [12]. It would be useful to decide which other fixed cofactor ranges have positive lower density, whether their natural densities exist, and how their overlaps affect the coverage of G_E .

FORMALIZATION

METHODOLOGY AND AI USAGE

This paper presents the results of a long-term large-scale collaborative project, lasting from November 2025 to August 2026. Its partial results were gradually announced and presented on the forum of the *Erdős Problems* website [2], which presents a reliable timeline of project’s development. The authors contributed with their thoughts, ideas, proofs, and formalizations, some of which were assisted by generative AI tools. More specifically,

- E.H. proved the adaptation of the Luca–Pomerance collision sieve to uniformly abundant witnesses in the proof of Theorem 1.2 (Lemmas 4.6 and 4.7 and Proposition 4.8), the exact forced modulus of Lemma 2.2, and the periodic envelope V_A with its density bounds (Proposition 5.2 and Corollary 5.3). The proofs of Proposition 6.1, the complement identity (6.8) with the lower bound for η_A , Proposition 6.6 with its corollary, and the prime-cofactor ceiling $\bar{d}(F_p(\mathbb{N})) \leq 1/p$ were generated with the help of Anthropic’s Claude Opus 5 and OpenAI’s GPT-5.6 Sol, audited by Claude Fable 5 and GPT-6 Astra, and then verified by E.H. The same models were used for polishing the writeup of E.H.’s parts and for numerical checks.
- Principia Math’s autonomous research harness, using the models GPT-5.5 and Claude Opus 4.8, resolved the limsup question in Erdős Problem 1054, proving the limsup variant of Theorem 1.3. The accompanying Lean formalization includes the circle-method proof of the almost-all binary Goldbach theorem used as an analytic input. The almost-all Goldbach result was initially useful, but was subsequently made redundant by the proof every integer besides 2 and 5 is representable. The formalization of this result, due to H.C. is also included in the Lean formalization repository.

The authors independently verified the AI-assisted proofs, carefully inspected the AI-generated parts of the text, rewrote them as appropriate, and they assume full responsibility for the final manuscript. Furthermore, OpenAI’s ChatGPT-6 Astra was used for several rounds of copyediting and proofreading.

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